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## 1. Operations using Data Types and Operators

### 1.1 Evaluate expressions to identify the data types Python assigns tovariables

• str, int, float, and bool

**1.**

Consider the following python code:

1) `age=0`
2) `minor=False`
3) `name='Durga'`

The types of age,minor and name variables respectively:

A) int, bool, str
B) bool, bool, str
C) int, bool, char
D) float, bool, str

**2.**

Consider the following python code:

```
weight = 62.4
zip='880098'
value=+23E4
```

The types of weight, zip and value variables respectively:
A) float, str, str
B) int, str, float
C) double, str, float
D) float, str, float

**3.**

You create the following program to locate a conference room and display the room name. Line numbers are included for reference only.

```
01 rooms = {1: 'Left Conference Room', 2: 'Right conference Room'}
02 room = input('Enter the room number: ')
03 if not room in rooms:
04   print('Room does not exist.')
05 else:
06   print("The room name is " + rooms[room])
```

Colleagues report that the program sometimes produces incorrect results. You need to troubleshoot the program. Use the drop-down menus to select the answer choice mat answers each question based on the information presented in the code segment.

Which two data types are stored in the rooms list at line 01? [1] .

- bool and string
- float and bool
- int and string
- float and int

What is the data type of room at line 02? [2] .

- bool
- float
- int
- string

Why does line 03 fail to find the rooms? [3] .

- Invalid syntax
- Mismatched data type(s)
- Misnamed variable(s)

## 4.

You are writing a program that calculates a user's year of birth. The program asks users for their age and the current year, then outputs the user's year of birth. You write the following code. Line numbers are included for reference only.

```
01 age = input(”Enter your age: “)
02 year = input(“Enter the four digit year: ”)
03 born = eval(year) - eval(age)
04 message = “You were born in “ + str(born)
05 print(message)
```

You need to ensure that the program uses the appropriate data types. What data types are used? Use the drop-down menus to select the answer choice that answers each question based on the information presented in ¡ne code segment.

What data type is age in line 01? int / str / float / bool

What data type is born in line 03? int / str / float / bool

What data type is message in line 04? int / str / float / bool

## 5.

During school holidays, you volunteer to explain some basic programming concepts to your younger siblings. You want to introduce the concept of data types in Python. You create the following three code segments:

```
# Code segment 1
x1 = "20"
y1 = 3
a = x1 * y1
# Code segment 2
x2 = 6
y2 = 4
b = x2 / y2
# Code segment 3
x3 = 2.5
y3 = 1
c = x3 / y3
```

You need to evaluate the code segments.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.

| | Yes | No |
|---|---|---|
| After executing code segment 1, the data type of variable a is str. | ○ | ○ |
| After executing code segment 2, the data type of variable b is float. | ○ | ○ |
| After executing code segment 3, the data type of variable c is int. | ○ | ○ |

**6.**

Match the data type to the type operations. To answer, drag the appropriate data type to the correct type operation. Each data type may be used once, more than once, or not at all.

**Data Types**

| int | float | str | bool |
|---|---|---|---|

**Answer Area**

| | |
|---|---|
| type (+1E10) | |
| type (5.0) | |
| type ("True") | |
| type (False) | |

**7.**

You are writing a Python program. The program collects customer data and stores it in a database. The program handles a wide variety of data. You need to ensure that the program handles the data correctly so that it can be stored in the database correctly.Match the data type to the code segment. To answer, drag the appropriate data type from the column on the left to its code segment on the right. Each data type may be used once, more than once, or not at all.

**Operations**

| bool | float | int | str |
|---|---|---|---|

**Answer Area**

| | |
|---|---|
| | age = 2 |
| | minor = False |
| | name = "Contoso" |
| | weight = 123.5 |
| | zip = "81000" |

**8.**

Consider the Variable declarations:

```
a='5'
b='2'
```

Which of the following expressions are of type str

A) a+b
B) a*b
C) a-b
D) a*2

Answer: A and D

**9.**

You are writing a Python program. You required to handle data types properly.
Consider the code segment:

1) `a=10+20`
2) `b='10'+'20'`
3) `c='10'*3`

Identify the types of a,b and c?

A) a is of int type,b is of str type and c is of str type
B) a is of int type,b is of str type and c is of int type
C) a is of int type,b is of int type and c is of int type
D) a is of int type ,b and c are invalid declarations

## 10.

You are writing a function that returns the data type of the value that is passed in. You write the following code. Line numbers are included for reference only.

```
01 def checkType(value):
02    dataType = type(value)
03    return dataType
04 print(checkType(True))
05 print(checkType(1.0))
06 print(checkType(1))
07 print(checkType(”True”))
```

Use the drop-down menus to select the answer choice that answers each question based on the information presented in the code segment.

What is printed at line 04? :

- <class 'bool'>
- <class 'int'>
- <class 'float'>
- <class 'str'>

What is printed at line 05? :

- <class 'bool'>
- <class 'int'>
- <class 'float'>
- <class 'str'>

What is printed at line 06? :

- <class 'bool'>
- <class 'int'>
- <class 'float'>
- <class 'str'>

What is printed at line 07? :

- <class 'bool'>
- <class 'int'>
- <class 'float'>
- <class 'str'>

## 1.2 Perform and analyze data and data type operations

• Data type conversion, indexing, slicing, construct data structures, lists, list operations (including sorting, merging, appending, inserting, removing, finding maximum and minimum, and reversing)

### 11.

You have the following code:

```
1) a=bool([False])
2) b=bool(3)
3) c=bool("")
4) d=bool(' ')
```

Which of the variables will represent False:
A) a
B) b
**C) c**
D) d

### 12.

Consider the following variable declarations:

```
1) a= bool([])
2) b= bool(())
3) c= bool(range(0))
4) d= bool({})
5) e= bool(set())
```

Which of the above variables represent True ?
A) c
B) a ,b, c, d
C) All Variables represent True
**D) None of the variables represents True**

### 13.

```
1) a=bool(0)
2) b=bool(3)
3) c=bool(0.5)
4) d=bool(0.0)
```

Which variables represent True?
A) a,b
B) b,c
C) c,d
D) d,a
E) All Variables

### 14.

You are writing a Python program to read two int values from the keyboard and print the sum.

```
1) x=input('Enter First Number:')
2) y=input('Enter Second Number:')
3) #Line-1
```

Which of the following code we have to write at Line-1 to print sum of given numbers?
A) print('The Result:'+(int(x)+int(y)))
B) print('The Result:'+(int(x+y)))
C) print('The Result:'+str(int(x)+int(y)))
D) print('The Result:'+str(int(x+y)))

## 15.

Consider the python code:

```
1) result=str(bool(1) + float(10)/float(2))
2) print(result)
```

What is the output?

A) SyntaxError
B) TypeError
C) 6
D) 6.0

Answer: D

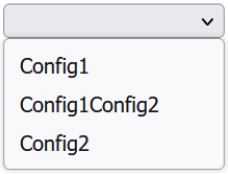
## 16.

You write the following code:

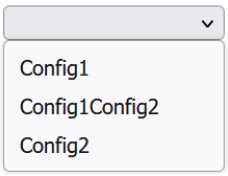
```
a = 'Config1'
print(a)
b = a
a += 'Config2'
print(a)
print(b)
```

Use the drop-down menus to select the answer choice that answers each question based on the information presented in the code segment.
Answer Area

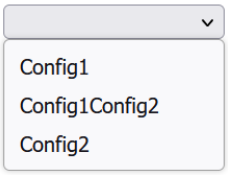
What is displayed after the first print? [1]




What is displayed after the second print? [2]




What is displayed after the third print? [3]




## 17.

You develop an application for your school. A list named colors contains 200 colors. You need to slice the list to display every other color starting with the second color.
Which code should you use?

A. colors[1:2]
B. colors [::2]
C. colors[2:2]
D. colors [1::2]

## 18.

You develop an application for your company. A list named employees contains 200 employee names, the last five being company management. You need to slice the list to display all employees excluding management. Which two code segments should you use? Each correct answer presents a complete solution. Choose two.

A. employees[0:-4]
B. employees [1:-5]
C. employees [:-5]
D. employees [0:-5]
E. employees [1:-4]

## 19.

You are developing a python application for your company. A list named employees contains 600 employee names,the last 3 being company management. You need to slice employees to display all employees excluding management. Which two code segments we should use?

A) employees[1:-2]
B) employees[:-3]
C) employees[1:-3]
D) employees[0:-2]
E) employees[0:-3]

## 20.

You are developing a python application for your company. A list named employees contains 500 employee names,the last 3 being company management. Which of the following represents only management employees.
A) employees[497:]
B) employees[-3:]
C) employees[497:500]
D) All the above

## 21.

You write the following code:

```
list_1 = [1, 2]
list_2 = [3, 4]
list_3 = list_1 + list_2
list_4 = list_3 * 3
print(list_4)
```

You run the code.
What is the output value?

A. [3, 6, 9, 12]
B. [1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4]
C. [[1, 2], [3, 4], [1, 2], [3, 4], [1, 2], [3, 4]]
D. [[1, 2, 3, 4], [1, 2, 3, 4], [1, 2, 3, 4]]

## 22.

Woodgrove Bank must generate a report that shows the average balance for all customers each day. The report must truncate the decimal portion of the balance. Which two code segments should you use? Each correct answer presents a complete solution. Choose two.

A. average_balance = total_deposits**number_of_customers
B. average_balance = total_deposits//number_of_customers
C. average_balance = int(total_deposits/number_of_customers)
D. average_balance = float(total_deposits//number_of_customers)

## 23.

You are creating a script to evaluate input and check for upper and lower case. Which four code segments should you use to develop the solution? To answer, move the appropriate code segment from the list of code segments to the answer area and arrange them in the correct order.

**Code Segments**

```
else:
  print(name, "is mixed case.")
```

```
else:
  print(name, "is lower case.")
```

```
name = input("Enter your name: ")
```

```
else:
  print(name, "is upper case.")
```

```
elif name.upper() = = name:
  print(name, "is all upper case.")
```

```
if name.lower() = = name:
  print(name, "is all lower case.")
```

**Answer Area**

## 24.

You are writing a Python program to perform arithmetic operations. You create the following code:

```
a = 11
b = 4
```

What is the result of each arithmetic expression? To answer, drag the appropriate expression from the column on the left to its result on the right. Each expression may be used once, more than once, or not at all.

**Results**

`print(a / b)` `print(a // b)` `print(a % b)`

**Answer Area**

| | |
|---|---|
| 2 | |
| 3 | |
| 2.75 | |

## 25.

You have the following list structure:

```
alph = "abcdefghijklnnopqrstuvwxyz"
```

You need to evaluate the result of performing various slicing operations. Match the result to the slicing operation. To answer, drag the appropriate result from the column on the left to its slicing operation on the right. Each result may be used once, more than once, or not at all.

| Operație | Rezultat |
|---|---|
| `[1] alph[3:15]` | |
| `[2] alph[3:15:3]` | |
| `[3] alph[3:15:-3]` | |
| `[4] alph[15:3:-3]` | |
| `[5] alph[15:3]` | |
| `[6] alph[::-3]` | |

A. zwtqnkheb
B. pmjg
C. defghijklmno
D. ponmlkjihgfe
E. defghijklmnop
F. dgjm
G. olif
H. ""

**26.**
What is a variable defined outside a function referred to as?
A) A static variable
B) A global variable
C) A local variable
D) An automatic variable

**27.**
What will be the output of the following Python code?

```
a = 10
b = 20

def change():
   global b
   a = 45
   b = 56

change()
print(a)
print(b)
```

A) 10,56
B) 45,56
C) 10, 20
D) Syntax Error

**28.**
What will be the output of the following Python code?

```
def change(i=1, j=2):
    i = i + j
    j = j + 1
    print(i, j)

change(j=1, i=2)
```

A) An exception is thrown because of conflicting values
B) 1 2
C) 3 3
D) 3 2

**29.**
What will be the output of the following python code?

```
names=['itvedant','Thane','Andheri','Navi Mumbai']
print(names[-1][-1])
```

A) Navi Mumbai
B) Mumbai
C) i
D) a

**30.**
What is the return type of the function id()?

A) int
B) float
C) bool
D) str

**31.**
What will be the output of the following python code?

```
f=lambda x:bool(x%2)
print(f(20), f(21))
```

A) False True
B) False False
C) True True
D) True False

**32.**
Consider the following lists:

```
n1=[10,20,30,40,50]
N2=[10,20,30,40,50]
print(n1 is n2)
print(n1 == n2)
```

What is the output?

A) False, False
B) True, True
C) True, False
D) False, True

**33.**
Consider the following lists:

```
1) n1=[10,20,30,40,50]
2) n2=[10,20,30,40,50]
3) print(n1 is n2)
4) print(n1 == n2)
5) n1=n2
6) print(n1 is n2)
7) print(n1 == n2)
```

What is the result?

A) False, False, True, True
B) False, True, False, True
C) False, True, True, True
D) True, False, True, False

## 34.

Consider the lists:

```
1) numbers=[10,20,30,40,50]
2) alphabets=['a','b','c','d','e']
3) print( numbers is alphabets)
4) print( numbers == alphabets)
5) numbers=alphabets
6) print( numbers is alphabets)
7) print( numbers == alphabets)
```

What is the result?

A) False, False, True, True
B) False, True, False, True
C) True, False, True, False
D) False, True, True, True

## 35.

Consider the code:

```
s='AB CD'
list=list(s)
list.append('EF')
print(list)
```

What is the result?
A) ['A', 'B', 'C', 'D', 'E', 'F']
B) {'A', 'B', '', 'C', 'D', 'EF'}
C) ['A', 'B', '', 'C', 'D', 'EF']
D) ('A', 'B', '', 'C', 'D', 'EF')

Answer: D

## 36.

Consider the code:

```
1) x='ACROTE'
2) y='APPLE'
3) z='TOMATO'
```

Which of the following won't print 'CAT' to the console

A) print(x[1]+y[0]+z[0])
B) print(x[2]+y[1]+z[1])
C) print(x[-5]+y[0]+z[0])
D) print(x[-5]+y[0]+z[-2])

Answer: B

## 37.

Consider the code:

```
1) s='Python is easy'
2) s1=s[-7:]
3) s2=s[-4:]
4) print(s1+s2)
```

What is the result?

A)is easyeasy
B)easyeasy
C)iseasyeasy
D)s easyeasy
E)is easy easy
Answer: A

## 38.

Consider the code:
1) s='Python is easy'
2) s1=s[6:-4]
3) #Line-1
4) print(len(s2))

To print 2 as output,which code we have to insert at Line-1

A)s2 = s1.lstrip()
B)s2 = s1.rstrip()
C)s2 = s1.lrstrip()
D)s2 = s1.strip()

Answer: D

## 39.

Consider the code:

```
t=([10,20],10,False)
```

Which line of the code assigns <class 'list'> to x

A) x= type(t)
B) x= type(t[0])
C) x= type(t[1])
D) x= type(t[0:])

Answer: B

## 40.

Consider the variable declaration

```
b = 'BANANA'
```

Which of the following lines will print 'AA' to the console?

A) print(b[1]+b[2])
B) print(b[1]+b[3])
C) print(b[1]+b[5])
D) print(b[3]+b[5])

Answer: B,C and D

## 41.

Consider the list:
list = ['Apple', 'Banana', 'Carrot', 'Mango']
Whitch of the following are the valid ways of accessing 'Mango':

Option A : list[3]
Option B: list[4]
Option C: list[-1]
Option D: list[0]

A) A and B
B) B and D
C) A and C
D) All options

## 42.

You are given the following Python code snippet:

```
x = 'TEXT'
```

Which line of code will assign the string 'TT' to the variable output?

A) output = x[1] + x[1]
B) output = x[1] + x[4]
C) output = x[0] + x[2]
D) output = x[0] + x[-1]

## 43.

Consider the Python code:

```
a=['a','b','c','d']
for i in a:
    a.append(i.upper())
print(a)
```

What is the result?
A) ['A', 'B', 'C', 'D']
B) ['a', 'b', 'c', 'd']
C) SyntaxError
D) MemoryError thrown at runtime

## 44.

Which of the following statements are valid? Choose all that apply

| | Valid | Invalid |
|---|---|---|
| Option A.<br>result = 456 + 456.0<br>type of result is int | ○ | ○ |
| Option B.<br>s = "Durga Sir's Python Classes are Good"<br>It causes an error because we can't use double quotes and single quotes | ○ | ○ |
| Option C.<br>The following line will print result: 4.5<br>print((7/2) + (False or True) + (9%3)) | ○ | ○ |
| Option D.<br>The following expression evaluates to 12<br>b = False+5-True+35//4 | ○ | ○ |

## 45.

Consider the code:

```
t=([10,20],10,False)
```

Whitch line of the code assigns <class 'list'> to x?

A) x = type(t)
B) x = type(t[0])
C) x=type(t[1])
D) x = type(t[0:1])

## 46.

Consider the code

```
1) count=input('Enter the number of customers of the bank:')
2) #Line-1
3) print(output)
```

Which code inserted at Line-1 will print 20 to the console if we pass 15 as count value from the console?

A)output=int(count)+5
B)output=count+5
C)output=str(count)+5
D)output=float(count)+5

Answer: A

## 47.

In which of the following cases we will get <class 'int'> as output?

| A) | B) | C) | D) |
|---|---|---|---|
| 1) x=47.0<br>2) print(type(x)) | 1) x='47'<br>2) print(type(x)) | 1) x=10+20j<br>2) print(type(x)) | 1) x=2**2**2<br>2) print(type(x)) |

Answer: D

## 48.

Which of the following are valid statements?

A) 5+False evaluates to False
B) True+1 evaluates to 2
C) True and False evaluates to False
D) True or False evaluates to False
E) type('') is <class 'bool'>

Answer: B,C

## 1.3 Determine the sequence of execution based on operator precedence

• Assignment (=, +=, -=, /=, %=, //=, **=), comparison (==, >=, <=, !=), logical (and, or, not), logical, arithmetic (+, -, /, //, %, **, unary + and -), identity (is), containment (in)

### 49.

Consider the code

1) `a=15`
2) `b=5`
3) `print(a/b)`

What is the result ?

A) 3
B) 3.0
C) 0
D) 0.0

### 50.

Consider the code

1) `a=21`
2) `b=6`
3) `print(a/b)`
4) `print(a//b)`
5) `print(a%b)`

What is the result?

A) 3 3 3
B) 3.5 3 3
C) 3.0 3 3
D) 3.5 3.5 3

### 51.

You have the following code:

1) `a=3`
2) `b=5`
3) `a += 2**3`
4) `a -=b//2//3`
5) `print(a)`

What is the result?

A)13
B)12
C)11
D)10

### 52.

Consider the following expression

1) `result=8//6%5+2**3-2`
2) `print(result)`

What is the result?

A) 6
B) 7
C) 8
D) 9

## 53.

Which of the following expression will generate max value?

A) `8%3*4`
B) `8-3*4`
C) `8//3*4`
D) `8/3*4`

Answer: D

## 54.

Which expression would evaluate to 2?

A) 3**2
B) 22%5
C) 13//4
D) 11/2
Answer: B

## 55.

Consider the code

```
a=7
b=3
c=5
d=1
```

Which line of the code assigns 9 to the output?

A) output=a%c+1
B) output=a+c//d
C) output=c*d-1
D) output=a+d*2

Answer: D

## 56.

Consider the code

1) x=3
2) x +=1
3) #Line-1

Which line should be inserted at Line-1 so that x value will become 16?

A)x+=2
B)x-=2
C)x*=2
D)x**=2

Answer: D

## 57.
Which expression evaluates to 4?

A) 7/2*3
B) 7%2+3
C) 7//2-3
D)7-2*3

Answer: B

## 58.
Consider the following expression:

```
6//4%5+2**3-2//3
```

This expression results to:
A) 9
B) 3
C) -1
D) 25

Answer: A

## 59.
Consider the code

```
1) x=2
2) y=6
3) x+=2**3
4) x//=y//2//3
5) print(x)
```

What is the output?
A)0
B)9
C)10
D)7

Answer: C

## 60.
Consider the Code

```
1) x=3/3+3**3-3
2) print(x)
```

What is the output?
A)25
B)32
C)0.11
D)25.0

Answer: D

## 61.

In which of the following cases, True will be printed to the console ?

| A | B | C | D | E |
|---|---|---|---|---|
| 1) a=45<br>2) b=45<br>3) print(a is not b) | 1) s1='Python'<br>2) s2='Python'.upper()<br>3) print(s1 is s2) | 1) x=[1,2,3]<br>2) y=[1,2,3]<br>3) print(x is y) | print('r' in 'durga') | print('is' in 'This IS a Fake News') |

## 62.

You evaluate the following code:

```
numList = [0,1,2,3,4]
print(5 in numList)
```

What is the output of the print statement?

A. 4
B. False
C. True
D. 5

## 63.

You develop an application for your company. You have the following code. Line numbers are ¡included for reference only.

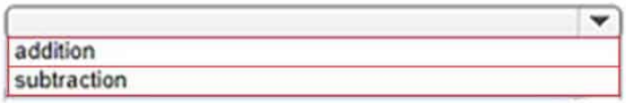
```
01 def main(a,b,c,d):
02     value = a+b*c-d
03     return value
```

Use the drop-down menus to select the answer choice that answers each question based on the information presented in the code segment.

Which part of the expression will be evaluated first? [1] .




Which operation will be evaluated second? [2] .




Which expression is equivalent to the expression in the function? [3] .




## 64.

Consider the expession:

```
result=a-b*c+d
```

Which of the following are valid?

**A) First b*c will be evaluated followed by subtraction and addition**
B) First b*c will be evaluated followed by addition and subtraction
C) First a-b will be evaluated followed by multiplication and addition
**D) The above expession is equivalent to a-(b*c)+d**

## 65.

Consider the code
1) `a=2`
2) `a += 1`
3) # Line-1

To make a value as 9,which expression required to place at Line-1
A) a*=2
B) a**=2
C) a+=2
D) a-=2

## 66.

Consider the python code
1) a=1
2) b=3
3) c=5
4) d=7

In Which of the following cases the result value is 0?
A) result = a+b*2
**B) result = a%b-1**
C) result = a-b//d
**D) result = a**d-1**

## 67.

In which of the following cases we will get same result
A) 23%5
B) 3**1
C) 11/3
D) 13//4

Answer: A,B,D

## 68.

Consider the code
1) a=1
2) b=2
3) c=4
4) d=6

Which of the following expression results -4?

A) (a+b)//c%d
B) (b+c)//a%d
C) (a+b)//c*d
D) (a+b)//d-c

Answer: D

## 69.

You are developing a Python application for your company.
You write the following code:

```
numList = [1,2,3,4,5]
alphaList = ["a","b","c","d","e"]
print(numList is alphaList)
print(numList == alphaList)
numList = alphaList
print(numList is alphaList)
print(numList == alphaList)
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

What is displayed after the first print? [True / False]

What is displayed after the second print? [True / False]

What is displayed after the third print? [True / False]

What is displayed after the fourth print? [True / False]

## 70.

```
1) subjects=['java','python','sap']
2) more_subjects=['java','python','sap']
3) extra_subjects=more_subjects
```

In which cases True will be printed to the console?

A) print(extra_subjects is more_subjects)
B) print(subjects is more_subjects)
C) print(subjects is extra_subjects)
D) print(subjects == extra_subjects)

Answer: A and D

## 71.

Consider the python code

```
1) numbers=[10,20,30,40]
2) x=0
```

In which of the following cases 10 will be printed to the console?

| A | B | C | D |
|---|---|---|---|
| 1) for i in (30,40,50):<br>2) if i in numbers:<br>3) x=x+5<br>4) print(x) | 1) for i in (30,40,50):<br>2) if i not in numbers:<br>3) x=x+5<br>4) print(x) | 1) for i in (30,40,50):<br>2) if i not in numbers:<br>3) x=x+10<br>4) print(x) | 1) for i in (30,40,50):<br>2) if i in numbers:<br>3) x=x+10<br>4) print(x) |

## 72.

Which of the following code snippet will produce the output:

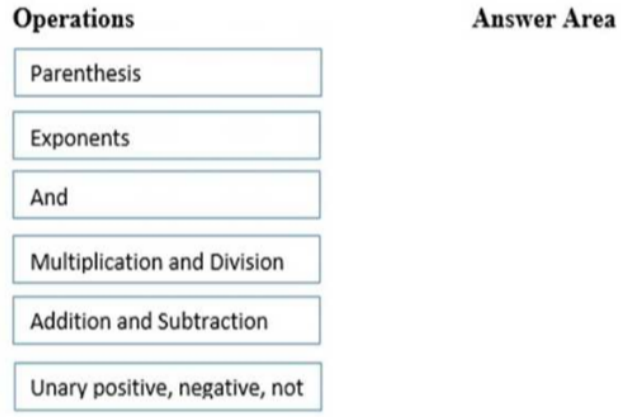
```
Boy
Cat
Dog
```

| A | B | C | D |
|---|---|---|---|
| 1) l=['Apple','Boy','Cat','Dog']<br>2) for x in l:<br>3) if len(x) == 3:<br>4) print(x) | 1) l=['Apple','Boy','Cat','Dog']<br>2) for x in l:<br>3) if len(x) != 3:<br>4) print(x) | 1) l=['Apple','Boy','Cat','Dog']<br>2) for x in l:<br>3) print(x) | 1) l=['Apple','Boy','Cat','Dog']<br>2) l1=l[1:]<br>3) for x in l1:<br>4) print(x) |

## 73.

The ABC company is converting an existing application to Python. You are creating documentation that will be used by several interns who are working on the team.

You need to ensure that arithmetic expressions are coded correctly. What is the correct order of operations for the six classes of operations ordered from first to last in order of precedence? To answer, move all operations from the list of operations to the answer area and arrange them in the correct order.




## 74.

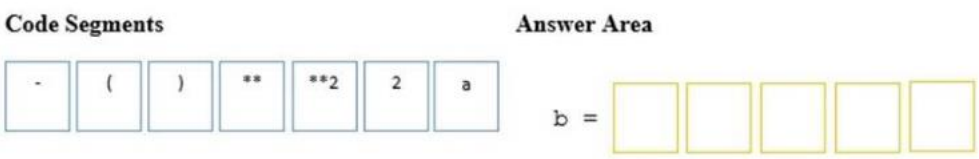
You are writing a Python program that evaluates an arithmetic formula. The formula is described as b equals a multiplied by negative one, then raised to the second power, where a is the value that will be input and b is the result.

You create the following code segment Line numbers are included for reference only.

```
01 a = eval(input("Enter a number for the equation: "))
02 b =
```

You need to ensure that the result is correct.

How should you complete the code on line 02? To answer, drag the appropriate code segment to the correct location. Each code segment may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.



## 75.

You are developing a Python program that compares numbers. You need to ensure that you are using the correct comparison operators. Evaluate each expression and indicate the correct result. To answer, drag the appropriate result from the column on the left to its expression on the right. Each result may be used once, more than once, or not at all.

[1] `0 or 5`
[2] `bool(0)`
[3] `None is None`
[4] `-5 < 0 < 5`

[1]: [2]: [3]: [4]:

True
False
5
None

## 76.

What will be the value of X in the following Python expression?

```
X = 2+9*((3*12)-8)/10
```

A) 30.0
B) 30.8
C) 28.4
D) 27.2

## 77.

Operators with the same precedence are evaluated in which manner?
A) Left to Right
B) Right to Left
C) Can't say
D) None of the mentioned

## 78.

You are writing a python program that evaluates an arithmetic expression.
The expression is described as b is equals a multiplied by negative one, then raised to the second power, where a is the value which will be input and b is result.

```
a=eval(input('Enter a number for the expression:'))
```

Which of the following is valid expression for the given requirement?

A) b = (a) ** -2
B) b = (-a) ** 2
C) b = (a-) ** 2
D) b = -(a) ** 2

## 79.

Which expression evaluates to 4?

A) 7//2-3
B) 7%2+3
C) 7/2*3
D) 7-2*3

## 1.4 Select operators to achieve the intended results

• Assignment (=, +=, -=, /=, %=, //=, **=), comparison (==, >=, <=, !=), logical (and, or, not), logical, arithmetic (+, -, /, //, %, **, unary + and -), identity (is), containment (in)

### 80.

Evaluate the following arithmetic expression:

```
(3*(1+2)**2 - (2**2)*3)
```

What is the result?

A. 3
B. 13
C. 15
D. 69

### 81.

Consider the following expression

```
result=(2*(3+4)**2-(3**3)*3)
```

What is result value?

A)17
B)16
C)18
D)19

### 82.

Which of the following expressions results in an error?

A) float(‘10’)
B) int(‘10’)
C) float(’10.8’)
D) int(’10.8’

### 83.

Consider the code

```
x='10'
y='20'
```

The type of x+y ?

A)int
B)float
C)str
D)complex

Answer: C

### 84.

Consider the code

```
a=float('123.456')
```

Which expression evaluates to 2?

A) int(a)+False
B) bool(a)+True
C) str(a)
D) bool(a)

Answer: B

## 85.

Consider the following code

```
1) x= 'Durga'
2) y= 'Durga'
3) result=condition
4) print(result)
```

For which of the following condition True will be printed to the console?

A) x is y
B) x is not y
C) x != y
D) x < y

Answer: A

## 86.

Consider the code:

```
1) x= 8
2) y= 10
3) result= x//3*3/2+y%2**2
4) print(result)
```

What is the result?
A) 5
B) 5.0
C) 6.0
D) 7.0

Answer: B

## 87.

Consider the Python code:

```
1) l1=['sunny','bunny','chinny','vinny']
2) l2=['sunny','bunny','chinny','vinny']
3) print(l1 is not l2)
4) print(l1 != l2)
5) l1=l2
6) print(l1 is not l2)
7) print(l1 != l2)
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| True<br>True<br>False<br>False | True<br>False<br>True<br>False | True<br>False<br>False<br>True | True<br>False<br>False<br>False |

Answer: D

## 88.

Consider the Python Code

```
1) l1=['sunny','bunny','chinny','vinny']
2) l2=['sunny','bunny','chinny','vinny']
3) print(l1 is l2)
4) print(l1 == l2)
5) l1=l2
6) print(l1 is l2)
7) print(l1 == l2)
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| False<br>True<br>True<br>True | False<br>False<br>True<br>True | False<br>True<br>False<br>True | False<br>True<br>True<br>False |

## 89.

Consider the python code:

```
1) print(10==10 and 20!=20)
2) print(10==10 or 20!=20)
3) print( not 10==10)
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| True<br>True<br>False | False<br>True<br>True | False<br>True<br>True | False<br>False<br>False |

Answer: C

## 90.

Consider the code:

```
1) print(not 0)
2) print(not 10)
3) print(not '')
4) print(not 'durga')
5) print(not None)
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| True<br>False<br>False<br>False<br>True | True<br>False<br>True<br>False<br>True | False<br>False<br>True<br>False<br>True | True<br>False<br>True<br>False<br>False |

Answer: B

## 91.

Consider the code:

```
1) lst = [7, 8, 9]
2) b = lst[:]
3) print(b is lst)
4) print(b == lst)
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| False<br>True | False<br>False | True<br>False | True<br>True |

Answer: A

## 2. Flow Control with Decisions and Loops

### 2.1 Construct and analyze code segments that use branching statements

• if, elif, else, nested and compound conditional expressions

### 93.

A classmate has asked you to debug the following code:

```
x = 4
  while x >= 1:
    if x % 4 == 0:
      print ("party")
    elif x - 2 < 0:
      print("cake")
    elif x / 3 == 0:
      print("greeting")
    else:
      print("birthday")
    x = x - 1
```

What is the output that is printed to the screen?

| A. | B. | C. | D. |
|---|---|---|---|
| birthday<br>party<br>greeting<br>cake | party<br>birthday<br>birthday<br>cake | party<br>greeting<br>birthday<br>cake | birthday<br>greeting<br>party<br>cake |

### 94.

You are writing a Python application for a dance studio. The studio wants to encourage youth and seniors to sign up. Minors and seniors must receive a 10% discount.
You write the following code. Line numbers are included for reference only.

```
01 def get_discount(minor, senior):
02 discount = .1
03
04    discount = 0
05 return discount
```

You need to complete the code. Which code should you add on line 03?

A. if not (minor and senior):
B. if not (minor or senior):
C. if (not minor) and senior:
D. if (not minor) or senior:

### 95.

You are creating a Python program that compares numbers.
You create the following code. Line numbers are included for reference only.

```
01 num1 = eval(input("Please enter the first number: "))
02 num2 = eval(input("Please enter the second number: "))
03 if num1 == num2:
04    print("The two numbers are equal. ")
05 if num1 <= num2:
06    print("Number 1 is less than number 2. ")
07 if num1 > num2:
08    print("Number 1 is greater than number 2. ")
09 if num2 = num1:
10    print("The two numbers are the same. ")
```

You need to ensure that the comparisons are accurate.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.

| | Yes | No |
|---|---|---|
| The print statement at line 04 will only print if the two numbers are equal in value. | ○ | ○ |
| The print statement at line 06 will only print if num1 is less than num2. | ○ | ○ |
| The print statement at line 08 will only print if num1 is greater than num2. | ○ | ○ |
| The statement at line 09 is an invalid comparison. | ○ | ○ |

## 96.

We are developing loan collection agent application. Consider the code:

```
1) collected_amount=3000
2) commission=0
3) if collected_amount <= 2000:
4)    commission=50
5) elif collected_amount> 2500 and collected_amount<3000:
6)    commission=100
7) elif collected_amount>2500:
8)    commission=150
9) if collected_amount>=3000:
10)   commission+=200
```

What will be the value of commission?

A) 350
B) 200
C) 150
D) 100

Answer:A

## 97.

You are developing online shopping application.
Consider the code:

```
01 order_value = 1500
02 state = 'ap'
03 delivery_charge = 0

04 if state in ['up', 'mp', 'ts']:
05    if order_value <= 1000:
06       delivery_charge = 50
07    elif 1000 < order_value < 2000:
08       delivery_charge = 100
09    else:
10        delivery_charge = 150
11 else:
12    delivery_charge = 25

13 if state in ['lp', 'kp', 'ap']:
14    if order_value > 1000:
15        delivery_charge += 20
16    if order_value < 2000 and state in ['kp', 'ap']:
17        delivery_charge += 30
18    else:
19        delivery_charge += 15
20 print(delivery_charge)
```

What is the result?

A) 65
B) 75
C) 85
D) 55

Answer: B

## 98.

You are designing a decision structure to convert a student's numeric grade to a letter grade. The program must assign a letter grade as specified in the following table:

| Percentage range | Letter grade |
|---|---|
| 90 through 100 | A |
| 80 through 89 | B |
| 70 through 79 | C |
| 65 through 69 | D |
| 0 through 64 | F |

For example, if the user enters a 90, the output should be, "Your letter grade is A." Likewise, if a user enters an 89, the output should be "Your letter grade is B."

How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
#Letter Grade converter
grade = int(input("Enter a numeric grade"))
[1]
    letter_grade = 'A'
[2]
    letter_grade = 'B'
[3]
    letter_grade = 'C'
[4]
    letter_grade = 'D'
else:
    letter_grade = 'F'
print("Your letter grade is : ", letter_grade)
```

[1]:
- if grade <= 90:
- if grade > 80:
- if grade > 70:
- if grade > 65:

[2]:
- if grade >= 90:
- if grade >= 80:
- if grade >= 70:
- if grade >= 65:

[3]:
- elif grade > 90:
- elif grade > 80:
- elif grade > 70:
- elif grade > 65:

[4]:
- elif grade >= 90:
- elif grade >= 80:
- elif grade >= 70:
- elif grade >= 65:

## 99.

You are writing a python script to convert student marks into grade. The grades are defined as follows:
90 through 100 --> A grade
80 through 89 -> B grade
70 through 79 --> C grade
65 through 69 --> D grade
0 through 64 --> E grade

And developed application is:

```
#Grade converter
marks=int(input("Enter Student marks:"))
if marks >=90:
    grade = 'A'
if marks >=80:
    grade = 'B'
if marks >=70:
    grade = 'C'
if marks >=65:
    grade = 'D'
else:
    grade = 'E'
print("Your grade is : ", grade)
```

Whitch of the following changes should be performed to fulfill the requirement?

A) Line 3 should be replaced with if marks <= 90:
B) Line 5 should be replaced with if marks>=80 and marks <= 90:
C) Line 7 should be replaced with if marks>=70 and marks <= 80:
D) No changes are required.

## 100.

We are developing one school automation application. If student marks between 80 and 100,then we have to offer 'A' grade. Which code block we have to use?

| A) | B) | C) | D) |
|---|---|---|---|
| 1) if marks>=80 and marks>=100:<br>2) grade='A' | 1) if marks>=80 or marks<=100:<br>2) grade='A' | 1) if 80<=marks<=100:<br>2) grade='A' | 1) if marks>80:<br>2) grade='A' |

Answer: C

## 101.

Consider the following code:

```
1) marks=[30,40,50,45,50,100]
2) average=sum(marks)//len(marks)
3) grades={1:'A',2:'B',3:'C',4:'D'}
4) if average>=90 and average<=100:
5)    key=1
6) elif average>=80 and average<90:
7)    key=2
8) elif average>=50 and average<80:
9)    key=3
10) else:
11)   key=4
12) print(grades[key])
```

Which grade will be printed to the console?

A) A
B) B
C) C
D) D

Answer: C

## 102.

You are coding a math utility by using Python. You are writing a function to compute roots. The function must meet the following requirements:

- If a is non-negative, return a**(1/b)
- If a is negative and even, return “Result is an imaginary number”
- If a is negative and odd, return -(-a)**(1/b)

How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
def safe_root(a, b):
   [1]
      answer = a**(1/b)
   [2]
      [3]
         answer = "Result is an imaginary number"
      [4]
         answer = -(-a)**(1/b)
   return answer
```

[1]: [2]: [3]: [4]:

if a >= 0:
if a % 2 == 0:
else:
elif:

## 103.

You are coding a math utility by using Python. You are writing a function to compute roots. The function must meet the following requirements:

- If a is non-negative, return a**(1/b)
- If a is negative and even, return “Result is an imaginary number”
- If a is negative and odd, return -(-a)**(1/b)

How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
def safe_root(a, b):
    [1]
        answer = a**(1/b)
    [2]
        [3]
            answer = "Result is an imaginary number"
        [4]
            answer = -(-a)**(1/b)
    return answer
```

[1]: [2]: [3]: [4]:

if a >= 0:
if a % 2 == 0:
else:
elif:

## 104.

You work for a company that distributes media for all ages.

You are writing a function that assigns a rating based on a user’s age. The function must meet the following requirements:

- Anyone 18 years old or older receives a rating of “A”.
- Anyone 13 or older, but younger than 18, receives a rating of “T’.
- Anyone 12 years old or younger receives a rating of “C”.
- If the age is unknown, the rating is set to “C’.

You need to complete the code to meet the requirements.

How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
def get_rating(age):
  rating = ""
  if [1]
  elif [2]
  elif [3]
  else [4]
  return rating
```

[1]: [2]: [3]: [4]:

age < 13 : rating = "C"
age < 18 : rating = "T"
:rating = "A"
age== None:rating = "C"

## 105.

You work for a company that distributes media for all ages.
You are writing a function that assigns a rating based on a user's age.The function must meet the following requirements.

- Anyone 18 years old or older receives a rating of "A"
- Anyone 13 or older,but younger than 18, receives a rating of "T"
- Anyone 12 years old or younger receives a rating of "C"
- If the age is unknown ,the rating is set to "C"

Which of the following code meets above requirements:

| A | B | C | D |
|---|---|---|---|
| def get_rating(age):<br>if age>=18:<br>rating="A"<br>elif age>=13:<br>rating="T"<br>else:<br>rating="C"<br>return rating | def get_rating(age):<br>if age>=18:<br>rating="A"<br>if age>=13:<br>rating="T"<br>else:<br>rating="C"<br>return rating | def get_rating(age):<br>if age>18:<br>rating="A"<br>elif age>13:<br>rating="T"<br>else:<br>rating="C"<br>return rating | def get_rating(age):<br>if age>=18:<br>rating="A"<br>elif age>=13:<br>rating="T"<br>else:<br>rating="C"<br>pass rating |

## 106.

Southridge Video needs a way to determine the cost that a customer will pay for renting a DVD. The cost is dependent on the time of day the DVD is returned. However, there are also special rates on Thursdays and Sundays. The fee structure is shown in the following list:

- The cost is $1.59 per night.
- If the DVD is returned after 8 PM, the customer will be charged an extra day.
- If the video is rented on a Sunday, the customer gets 30% off for as long as they keep the video.
- If the video is rented on a Thursday, the customer gets 50% off for as long as they keep the video.

You need to write code to meet the requirements.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
#Southridge Video, DVD Rental calculator
ontime = input("Was video returned before 8 pm? y or n").lower()
days_rented = int(input("How many days was video rented? "))
day_rented = input("What day was the video rented?").capitalize()
cost_per_day = 1.59
if ontime [1]
  days_rented +=1
if day_rented [2]
  total = (days_rented * cost_per_day) * .7
elif day_rented [3]
  total = (days_rented * cost_per_day) * .5
else:
  total = days_rented * cost_per_day
print("Cost of the END rental is : $", total)
```

[1]: != "n":, == "n":, == "y":

[2]: == "Sunday":, >= "Sunday":, is "Sunday":

[3]: == "Thursday":, <= "Thursday":, is "Thursday":

## 107.

The XYZ Book Company needs a way to determine the cost that a student will pay for renting a Book.
● The Cost is dependent on the time of the Book is returned.
● However there are also special rates on Saturday and Sundays.
● The Fee Structure is shown in the following list:
● The cost is $3.00 per night.
If the Book is returned after 9PM, the student will be charged an extra day. If the Book is rented on a Sunday,the student will get 50% off for as long as they the keep the book.
If the Book is rented on a Saturday,the student will get 30% off for as long as they keep the book.
We need to write the code to meet this requirements.

# XYZ Book Rented Amount Calculator

```
ontime = input("Was Book returned before 9 pm? y or n").lower()
days_rented = int(input("How many days was video rented? "))
day_rented = input("What day was the video rented?").capitalize()
cost_per_day = 3.00
if ontime == 'n'
  days_rented = days_rented + 1
if day_rented == 'Sunday'
  total = (days_rented * cost_per_day) *0.5
elif day_rented == 'Saturday'
  total = (days_rented * cost_per_day) *0.7
else:
  total = days_rented * cost_per_day
print("The Cost of the Book rental is : $", total)
```

If the Book rented on 'Sunday',the number of days Book rented is 5 and Book returned after 9PM then what is the result?
A) The Cost of Book Rental is: $7.0
B) The Cost of Book Rental is: $8.0
C) The Cost of Book Rental is: $9.0
D) The Cost of Book Rental is: $10.0

## 108.

We are developing gold loan application for XYZ company.

```
1) amount=float(input('Enter Loan Amount:'))
2) interest_rate=0
3) if amount > 0 and amount<= 50000:
4)    interest_rate = 10
5)
6) elif amount > 50000 and amount<100000:
7)    interest_rate = 12
8)
9) elif amount >= 100000 and amount<150000:
10)   interest_rate = 16
11)
12) else:
13)   interest_rate = 22
```

For which of the following user input interest_rate will be 12.
A) 50000
B) 50001
C) 100000
D) 100001
E) 150000

Answer: B

## 109.

You are writing a Python program to ask the user to enter a number and determine if the number is 1 digit 2 digits, or more than 2 digits long. You need to write the program.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
num = int(input("Enter a number with 1 or 2 digits: "))
digits = 0;
[1]
   digits = "1"
[2]
   digits = 2
[3]
   digits = ">2"
print(digits + " digits.")
```

:

[1]:
- if num > -10 and num < 10:
- if num > -100 and num < 100:

[2]:
- if num > -100 and num < 100:
- elif num > -100 and num < 100:
- if num > -10 and num < 10:
- elif num > -10 and num < 10:

[3]:
- else:
- elif:

## 110.

You are creating a function to calculate admission fees by using Python. Admission fees are calculated based on the following rules:

- Anyone under age 5 = free admission
- Anyone age 5 or older who is in school = 10 USD
- Anyone age 5 to 17 who is not in school = 20 USD
- Anyone older than age 17 who is not in school = 5O USD

How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
def admission_fee(age, school):
  rate = 0
  [1]
    rate = 10
  [2]
    [3]
      rate = 20
    else:
      rate = 50
  return rate
```

:

[1]:
- if age >= 5 and school == True:
- if age >= 5 and age <= 17:
- if age >= 5 and school == False:

[2]:
- elif age >= 5 and school == False:
- else age >= 5 and school == False:
- elif age >= 5 and school == True:

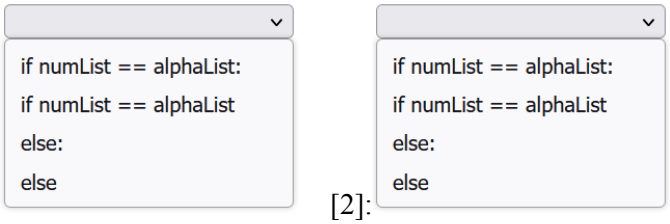
[3]:
- if age >= 5 and school == True:
- if age >= 5 and school == False:
- if age <= 17:

## 111.

You develop a Python application for your company. How should you complete the code so that the print statements are accurate? To answer, select the appropriate code segments in the answer area.

```
numList = [1,2,3,4,5]
alphaList = ["a","b","c","d","e"]
[1]
   print("The values in numList are equal to alphaList")
[2]
   print("The values in numList are not equal to alphaList")
```

[1]: [2]:




## 112.

Consider the Python code:

```
a=5
b=10
c=2
d=True

x=a+b*c
y=a+b/d

if(condition):
   print('Valid')
else:
   print('Invalid')
```

To print 'Valid' to the console, whitch condition we have to take for if statement?

A) x > y
B) x == y
C) x < y
D) x <= y

## 113.

We are developing a online shopping application. Consider the code:

```
d = input('Enter day of the week:')
discount_percentage = 3
if d == 'Monday':
   discount_percentage+=5
elif d == 'Tuesday':
   discount_percentage+=7
elif d == 'Saturday':
   discount_percentage+=10
elif d == 'Sunday':
   discount_percentage+=20
else:
   discount_percentage+=2
```

To get 5 as discount_percentage, whitch of the following input should be provided end user?

A) Monday
B) Tuesday
C) Thursday
D) Saturday
E) Sunday

Answer: C

## 114.
Which of the following is True about else block?

A) else block will be executed if there is no exception is try bloack
B) without writing except block we can't write else block
C) for the same try we can write atmost one else block
D) All the above

Answer : D

## 115.
We are developing an application for leave approval in XYZ Company.

```
days=int(input('Enter number of days for leave:')) cause=input('Enter the cause:')
if days==1:
   print('Leave will be approved immediately')
elif days>1 and days<=3:
   if cause=='Sick":
      print('Leave will be approved immediately')
   else:
      print('Needs Lead Approval')
elif days>3 and days<5:
   if cause=='Sick':
      print('Needs Manager Approval')
   else:
      print("Needs Director Approval')
elif days>=5 and days<=10:
   print('Needs Director Approval')
```

In which of the following cases 'Needs Director Approval' will be printed to the console?
A) days = 2 and cause='sick'
B) days = 3 and cause='personal'
C) days = 4 and cause='sick'
D) days = 4 and cause='official'

Answer: D

## 116.
Consider the code

```
a=12
b=4
s='He shall not be happy if he does not work'
```

In which of the following cases result value will become 9
A) result=3 if None else a/b
B) result=s.find('not') if s else None
C) result=s.rfind('not') if s else None
D) result=5 if len(s)>4 else 6

Answer: B

## 2.2 Construct and analyze code segments that perform iteration

• while, for, break, continue, pass, nested loops, loops that include compound conditional expressions

### 117.

You find errors while evaluating the following code. Line numbers are included for reference only.

```
01 numbers = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
02 index = 0
03 while (index < 10)
04     print(numbers[index])
05
06     if numbers(index) = 6
07       break
08     else:
09       index += 1
```

You need to correct the code at line 03 and line 06.

How should you correct the code? Use the drop-down menus to select the answer choice that answers each question based on the information presented in the code segment.

Which code segment should you use at line 03?

- while (index < 10) :
- while [index < 10]
- while (index < 5) :
- while [index < 5]

Which code segment should you use at line 06?

- if numbers[index] == 6
- if numbers[index] == 6 :
- if numbers(index) = 6 :
- if numbers(index) != 6

### 118.

You are building a Python program that displays all of the prime numbers from 2 to 100.

How should you complete the code? To answer, drag the appropriate code segments to the correct location. Each code segment may be used once, more than once, or not at all.

You may need to drag the split bar between panes or scroll to view content.

**Code Segments**

```
p = 2
while p <= 100:
      is_prime = True
```

```
p = 2
is_prime = True
while p <= 100:
```

```
break
```

```
continue
```

```
p = p + 1
```

```
for i in range(2, p):
 if p / i = = 0:
     is_prime = False
```

```
for i in range(2, p):
 if p % i = = 0:
     is_prime = False
```

**Answer Area**

```
if is_prime = = True:
    print(p)
```

## 119.

You are creating a Python program that will let a user guess a number from 1 to 10. The user is allowed up to three guesses. You write the following code. Line numbers are included for reference only.

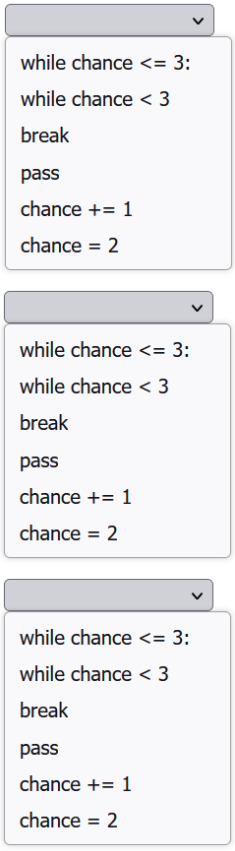
```
01 from random import randint
02 target = randint(1, 10)
03 chance = 1
04 print("Guess an integer from 1 to 10. You will have 3 chances.")
05
06 guess = int(input("Guess an integer: "))
07 if guess > target:
08     print("Guess is too high")
09 elif guess < target:
10     print("Guess is too low")
11 else:
12     print("Guess is just right!")
13
14
```

The program must allow three guesses. If the user guesses the correct number, the program must stop asking for guesses. How should you complete lines 05, 13, and 14? To answer, drag the appropriate code segments to the correct locations. Each code segment may be used once, more than once, or not at all.
You may need to drag the split bar between panes or scroll to view content

1. Which code segement should you use at line 05?

2. Which code segement should you use at line 13?

3. Which code segement should you use at line 05?



## 120.

You are developing a Python application for an online product distribution company. You need the program to iterate through a list of products and escape when a target product ID is found.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
productIdList = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
index = 0
[1]  (index < 10) :
    print(productIdList[index])
    if productIdList[index) == 6 :
        [2]
    else :
        [3]
```

[1]: while / for / if / break

[2]: while / for / if / break

[3]: continue / break / index += 1 / index = 1

## 121.

Luceme Publishing Company needs a way to find the count of particular letters in their publications to ensure that there is a good balance. It seems that there have been complaints about overuse of the letter e. You need to create a function to meet the requirements.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
#Function accepts list of words from a file,
#and letter to search for.
#Returns count of a particular letter in that list.
def count_letter(letter, word_list):
  count=0
  for [1]
    if [2]
      count += 1
  return count
word_list =[]
#word_list is populated a from file, code not shown.
letter = input("which letter would you like to count")
letter_count= count_letter(letter, word_list)
print("There are: " letter_count, " instances of " + letter)
```

[1]: word_list in word: / Word in word_list: / Word == word_list: / Word is word_list:

[2]: Word is letter: / Letter is word: / Word in letter: / Letter in word:

## 122.

Consider the following Python Code:

```
1) def count_letter(letter,word_list):
2)   count=0
3)   for word in word_list:
4)     if letter in word:
5)       count +=1
6)   return count
7) word_list=['apple','pears','orange','mango']
8) letter=input('Enter some alphabet symbol:')
9) letter_count=count_letter(letter,word_list)
10) print(letter_count)
```

If the user provides input 'a' then what is the result?
A) 1
B) 2
C) 3
D) 4

Answer: D

## 123.

Best For You Organics Company needs a simple program that their call center will use to enter survey data for a new coffee variety. The program must accept input and return the average rating based on a five-star scale.
The output must be rounded to two decimal places. You need to complete the code to meet the requirements.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
sum = count = done = 0
average = 0.0
while (done != -1):
   rating = [1] [ ]
   if rating == -1:
      break
   sum+= rating
   count+=1
average = float(sum/count)
[2] [ ] + [3] [ ]
```

[1]:
- print("Enter next rating (1-5), -1 for done")
- float(input("Enter next rating (1-5), -1 for done"))
- input("Enter next rating (1-5), -1 for done")
- input "Enter next rating (1-5), -1 for done")

[2]:
- output("The average star rating for NetVerZleep coffee is: "
- console.input("The average star rating for the new coffee is: "
- printline("The average star rating for the new coffee is: "
- print("The average star rating for the new coffee is: "

[3]:
- format(average, '.2f'))
- format(average, '.2d'))
- {average, '.2f'}
- format.average .{2d}

## 124.

The XYZ organics company needs a simple program that their call center will use to enter survey data for a new coffee variety. The program must accept input and return the average rating based on a five-star scale.The output must be rounded to two decimal places.
Consider the code:

```
1) sum=count=done=0
2) average=0.0
3) while(done != -1):
4)   rating=float(input('Enter Next Rating(1-5),-1 for done'))
5)   if rating == -1:
6)     break
7)   sum+=rating
8)   count+=1
9) average=float(sum/count)
10) #Line-1
```

Which of the following print() statement should be placed at Line-1 to meet requirement?

A) print('The average star rating for the new coffee is:{:.2f}'.format(average))
B) print('The average star rating for the new coffee is:{:.2d}'.format(average))
C) print('The average star rating for the new coffee is:{:2f}'.format(average))
D) print('The average star rating for the new coffee is:{:2.2d}'.format(average))

Answer: A

## 125.

Consider the code:

```
1) l=[10,20,[30,40],[50,60]]
2) count=0
3) for i in range(len(l)):
4)   if type(l[i])==list:
5)     count=count+1
6) print(count)
```

What is the result?
A) 1
B) 2
C) 3
D) 4

Answer : B

## 126.

Consider the code:

```
1) l=[10,(20,),{30},{},{},[40,50]]
2) count=0
3) for i in range(len(l)):
4)   if type(l[i])==list:
5)     count+=1
6)   elif type(l[i])==tuple:
7)     count+=2
8)   elif type(l[i])==set:
9)     count+=3
10)  elif type(l[i])==dict:
11)    count+=4
12)  else:
13)    count+=5
14) print(count)
```

What is the result?
A)17
B)18
C)19
D)20

Answer: C

## 127.

Consider the code:

```
1) t = (2,4,6,8,10,12)
2) d = {1:'A',2:'B',3:'C',4:'D',5:'E',6:'F'}
3) result=1
4) for t1 in t:
5)   if t1 in d:
6)     result+=t1
7) print(result)
```

What is the result?

A) 12
B) 13
C) 19
D) 6

Answer: B

## 128.

Consider the code:

```
1) t = (2,4,6,8,10,12)
2) d = {1:'A',2:'B',3:'C',4:'D',5:'E',6:'F'}
3) result=1
4) for t1 in t:
5)   if t1 in d:
6)     continue
7)   else:
8)     result+=t1
9) print(result)
```

What is the result?

A) 29
B) 30
C) 31
D) 32

Answer: C

## 129.

Consider the code:

```
1) values = [[3, 4, 5, 1], [33, 6, 1, 2]]
2)
3) v = values[0][0]
4) for lst in values:
5)   for element in lst:
6)     if v > element:
7)       v = element
8)
9) print(v)
```

What is the result?

A) 3
B) 2
C) 1
D) 4

Answer: C

## 130.

Tailspin Toys is building a basketball court for its employees to improve company morale. You are creating a Python program that employees can use to keep track of their average score. The program must allow users to enter their name and current scores. The program will output the user name and the users average score. The output must meet the following requirements:

- The user name must be left-aligned.
- If the user name has fewer than 20 characters, additional space must be added to the right.
- The average score must have three places to the left of the decimal point and one place to the right of the decimal (XXX.X).

How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
name = input("What is your name? ")
score = 0
count = 0
sum = 0
while(score != -1):
   score = int(input("Enter your scores: (-1 to end) "))
   if score == -1:
      break
   sum += score
   count += 1
average_score = sum / count
print("[1]      , your average score is: [2]      " %(name, average))
```

[1]: %-20i / %-20d / %-20f / %-20s

[2]: %1.4s / %4.1f / %4.1s / %1.4f

## 131.

Relecloud Virtual Learning asks you to debug some code that is causing problems with their payroll. They ask you to find the source of the payroll errors. The following variables have been declared:

```
employee_pay = [15000, 120000, 35000, 45000]
count = 0
sum = 0
```

There are two errors in the following code:

```
for index in range(0, len(employee_pay)-1)):
   count += 1
   sum += employee_pay[index]
average = sum//count
print("The total payroll is: ", sum )
print("The average salary is: ", average)
```

Which code should you use to fix the errors? To answer, select the appropriate code segments in the answer area.

```
for index in range [1]
   count += 1
   sum += employee_pay[index]
average = [2]
print("The total payroll is: ", sum )
print("The average salary is: ", average)
```

[1]: (size(employee_pay)): / (size(employee_pay)-1): / (len(employee_pay)+1): / (len(employee_pay)):

[2]: sum/count / sum**count / sum*count

## 132.

Adventure Works Cycles sales are so exceptional that they decide to give a bonus to all employees who do not make more than S150,000. The following formula applies to each employee based on their base salary and a flat bonus:
**New salary = current salary x 3% + a $500 bonus.**
You write code that reads the employee salaries into a variable named salary_list.
You need to complete the code that applies an increase to each eligible employee's salary.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
#Each salary in the list is updated based on increase. Employees making.
#$150,000 or more will not get a raise.
#Salary list is populated from employee database, code not shown.
[1]
   if salary_list[index] >= 150000:
      [2]
      salary_list[index] = (salary_list[index] * 1.03) + 500
```

[1]:
- for index in range(len(salary_list)+1):
- for index in range(len(salary_list)-1):
- for index in range(len(salary_list)):
- for index in salary_list:

[2]:
- exit()
- continue
- break
- end

## 133.

What will be the output of the following Python code?

```
True = False
while True:
    print(True)
    break
```

A. True
B. False
C. None
D. none of the mentioned

## 134.

What will be the output of the following Python code?

```
x=123
for i in x:
  print(i)
```

a) 1 2 3
b) 123
c) error
d) none of the mentioned

## 135.

What will be the output of the following Python code?

```
for i in range(5):
    if i == 5:
        break
    else:
        print(i)
else:
    print("Here")
```

A) 0 1 2 3 4 Here
B) 0 1 2 3 4 5 Here
C) 0 1 2 3 4
D) 1 2 3 4 5

## 136.

What will be the output of the following Python code?

```
for i in range(0):
print(i)
```

A) 0
B) no output
C) error
D) None of the mentioned.

## 137.

What will be the output of the following python code.

```
a = [0, 1, 2, 3]
i = -2
for i not in a:
   print(i)
   i += 1
```

A) -2 -1
B) 0
C) error
D) none of the mentioned

## 138.

What will be the output of the following Python code?

```
x = 1567
for i in x:
   print(i)
```

A) 1 5 6 7
B) 1567
C) error
D) none of the mentioned

## 139.

Consider the code:

```
l=[10, (20,), {30}, {}, {}, [48,50]]
count=0
for i in range(len(1)):
   if type(1[i])==list:
      count+=1
   elif type(1[i])==tuple:
      count+=2
   elif type(1[i])==set:
      count+=3
   elif type(1[i])==dict:
      count+=4
   else:
      count+=5
print(count)
```

What is the result?
A) 17
B) 18
C) 19
D) 20

## 140.

You are writing a python program that displays all prime numbers from 2 to 200.
Which of the following is the proper code to fulfilling our requirement?

| Code A | Code B | Code C | Code D |
|---|---|---|---|
| p = 2<br>is_prime = True<br>while p <= 200:<br>  for i in range(2, p):<br>  if p % i == 0:<br>  is_prime = False<br>  break<br>  if is_prime == True:<br>  print(p)<br>  p = p + 1 | p = 2<br>while p <= 200:<br>  is_prime = True<br>  for i in range(2, p):<br>  if p % i == 0:<br>  is_prime = False<br>  break<br>  if is_prime == True:<br>  print(p)<br>  p = p + 1 | p = 2<br>while p <= 200:<br>  is_prime = True<br>  for i in range(2, p):<br>  if p % i == 0:<br>  is_prime = False<br>  break<br>  if is_prime == False:<br>  print(p)<br>  p = p + 1 | p = 2<br>while p <= 200:<br>  is_prime = True<br>  for i in range(2, p):<br>  if p % i == 0:<br>  is_prime = False<br>  break<br>  if is_prime == True:<br>  print(p) |

A) Code A
B) Code B
C) Code C
D) Code D

## 3. Input and Output Operations

### 3.1 Construct and analyze code segments that perform file input and output operations

• open, close, read, write, append, check existence, delete, with statement

### 141.

You are writing a program to automate inventory. Your first task is to read a file of inventory transactions. The file contains sales from the previous day, including the item id, price, and quantity.
The following shows a sample of data from the file:
10, 200, 5
20, 100, 1
The code must meet the following requirements:

- Each line of the file must be read and printed
- If a blank line is encountered, it must be ignored
- When all lines have been read, the file must be closed

You create the following code. Line numbers are included for reference only.

```
01 inventory = open("inventory.txt", 'r')
02 eof = False
03 while eof == False:
04   line = inventory.readline()
05
06
07       print(line)
08     else:
09       print("End of file")
10       eof = True
11 inventory.close()
```

Which code should you write for line 05 and line 06?

| A. | B. | C. | D. |
|---|---|---|---|
| 05 if line != '':<br>06 if line !="": | 05 if line != '\n':<br>06 if line != None: | 05 if line != '\n':<br>06 if line != "": | 05 if line != '':<br>06 if line != "\n": |

### 142.

We are creating a function that reads a data file and prints each line of that file. Consider the following code:

```
1) import os
2) def read_file(file):
3)   line=None
4)   if os.path.isfile(file):
5)     data=open(file,'r')
6)   while line != '':
7)     line=data.readline()
8)     print(line)
```

The code attempts to read the file even if the file does not exist. You need to correct the code. which lines having identation problems?

A) First 3 Lines inside function
B) Last 3 Lines inside function
C) Last 2 Lines inside function
D) There is no indentation problem

Answer: B:

```
1) import os
2) def read_file(file):
3)   line=None
4)   if os.path.isfile(file):
5)     data=open(file,'r')
6)     while line != '':
7)       line=data.readline()
8)       print(line)
```

## 143.

You develop a Python application for your school. You need to read and write data to a text file. If the file does not exist it must be created. If the file has content the content must be removed.
Which code should you use?

A. open("local_data", "r+")
B. open("local_data", "w+")
C. open("local_data", "r")
D. open("local_data", "w")

## 144.

You develop a python application for your school. You need to read and write data to a text file. If the file does not exist, it must be created. If the file has the content, the content must be removed. Which code we have to use?

A) open('abc.txt', 'w+')
B) open('abc.txt', 'r')
C) open('abc.txt', 'a')
D) open('abc.txt', 'r+')

## 145.

You are creating a function that reads a data file and prints each line of the file. You write the following code. Line numbers are included for reference only.

```
01 import os
02 def read_file(file):
03     line = None
04     if os.path.isfile(file):
05         data = open(file, 'r')
06         while line != '':
07             line = data.readline()
08             print(line)
```

The code attempts to read the file even if the file does not exist. You need to correct the code. Which three lines have indentation problems? Each correct answer presents part of the solution. Choose three.

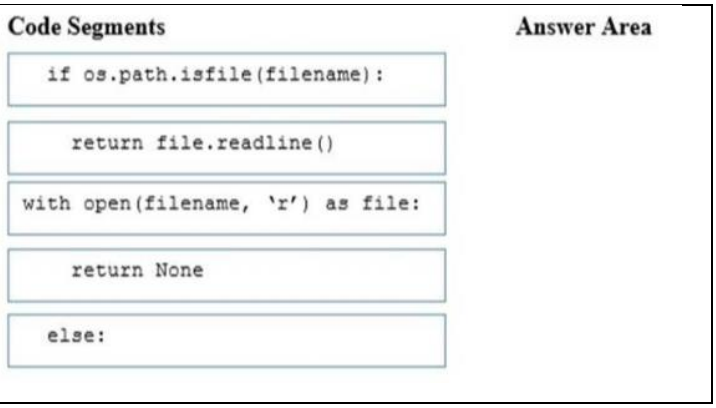
A. Line 01
B. Line 02
C. Line 03
D. Line 04
E. Line 05
F. Line 06
G. Line 07
H. Line 08

## 146.

You are writing a function that works with files. You need to ensure that the function returns None if the file does not exist. If the file does exist, the function must return the first line. You write the following code:

```
import os
def get_first_line(filename, mode):
```

In which order should you arrange the code segments to complete the function? To answer, move all code segments from the list of code segments to the answer area and arrange them in the correct order.



## 147.

Consider the following code.

```
1) import os
2) def get_data(filename,mode):
3)   if os.path.isfile(filename):
4)     with open(filename,'r') as file:
5)       return file.readline()
6)   else:
7)     return None
```

Which of the following are valid about this code?

A) This function returns the first line of the file if it is available
B) This function returns None if the file does not exist
C) This function returns total data present in the file
D) This function returns last line of the file

Answer: A and B

## 148.

Northwind Electric Cars needs help updating their file system using Python code. You must create a simple file manipulation program that performs the following actions:

- Checks to see if a file exists.
- If the file exists, displays its contents.
- If the file does not exist, creates a file using the specified name.
- Appends the phrase, "End of listing" to the file.

You need to complete the code to meet the requirements.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
import os
if [1]
   file = open('myFile.txt')
[2]
   file.close()
file= [3]
[4]  ("End of listing")
```

[1]: isfile('myFile.txt'): / os.exist('myFile.txt'): / os.find('myFile.txt'): / os.path.isfIIe('myFiIe.txt'):

[2]: output('myFile.txt') / print(file.get('myFile.txt')) / print(fiIe.read()) / print('myFile.txt')

[3]: open('myFile.txt', 'a') / open('myFile.txt', 'a+') / open('myFile.txt', 'w') / open('myFile.txt', 'w+')

[4]: append / file.add / file.write / write

## 149.

Which of the following statements are true?

A) When you open a file for reading, if the file does not exist, an error occurs
B) When you open a file for writing, if the file does not exist, a new file is created
C) When you open a file for writing, if the file exists, the existing file is overwritten with the new file
D) All of the mentioned

## 150.
To read the entire remaining contents of the file as a string from a file object infile, we use ____
A) infile.read(2)
B) infile.read()
C) infile.readline()
D) infile.readlines()

## 151.
What is the use of tell() method in python?
A) tells you the current position within the file
B) tells you the end position within the file
C) tells you the file is opened or not
D) none of the mentioned

## 152.
Which of the following string declarations spans more than one line and considers whitespace properly when the string is printed to the console?

| A) | B) | C) | D) |
|---|---|---|---|
| 1) s1='durga<br>2) software<br>3) solutions' | 1) s1="durga<br>2) software<br>3) solutions" | 1) s1='durga\n<br>2) software\n<br>3) solutions' | 1) s1='''durga<br>2) software<br>3) solutions''' |

Answer: D

## 153.
Consider the following code:

```
print(type(input('Enter some value:')))
```

if we enter 10 and 10.0 individually for every run what is the output?

| A) | B) | C) | D) |
|---|---|---|---|
| 1) <class 'str'><br>2) <class 'str'> | 1) <class 'int'><br>2) <class 'float'> | 1) <class 'int'><br>2) <class 'int'> | 1) <class 'float'><br>2) <class 'float'> |

Answer: A

## 154.
Consider the following code:

```
print(type(eval(input('Enter some value:'))))
```

if we enter 10 and 10.0 individually for every run what is the output?

| A) | B) | C) | D) |
|---|---|---|---|
| 1) <class 'str'><br>2) <class 'str'> | 1) <class 'int'><br>2) <class 'float'> | 1) <class 'int'><br>2) <class 'int'> | 1) <class 'float'><br>2) <class 'float'> |

Answer: B

## 155.

Which of the following statements are valid?

| | Valid | Invalid |
|---|---|---|
| A) s="Durga Sir's Python Classes are Good"<br>It causes error because we cannot use double quotes and single quotes simultaneously | ○ | ○ |
| B) result=456+456.0<br>type of result is int | ○ | ○ |
| C) The followng expression evaluates to 12<br>b=False+5-True+35//4 | ○ | ○ |
| D) The following line will print result:4.5<br>print('result:',(7/2)+(False or True)+(9%3)) | ○ | ○ |

Answer: C,D

## 156.

Conside the file abc.txt:

abc.txt:
---
```
Durga:10
Ravi:20
Shiva:30
Pavan:40
```

Consider the python code which is present in the same location of the file
test.py:
---
```
1) values=0
2) try:
3)   f=open('abc.txt','r')
4)   content=f.readlines()
5)   for line in content:
6)     values+=float(line.split(':')[1])
7)   f.close()
8) except Exception:
9)   print('Unable to open the file')
10) print(values)
```

What is the result?
A) Unable to open the file
B) 100
C) 100.0
D) 10.0

Answer: C

## 157.

To write 'Python Certificaton' to abc.txt file, which of the following is valid code?

| A) | B) | C) | D) |
|---|---|---|---|
| f=open('abc.txt','b')<br>f.write('Python Certificaton')<br>f.close() | f=open('abc.txt','r')<br>f.write('Python Certificaton')<br>f.close() | f=open('abc.txt')<br>f.write('Python Certificaton')<br>f.close() | f=open('abc.txt','w')<br>f.write('Python Certificaton')<br>f.close() |

Answer: D

## 158.

Consider the data present in the file: abc.txt

```
DURGASOFT,50,60,70,80,90
MICROSOFT,10,20,30,40,50
```

Which of the following is valid code to read total data from the file?

| A) | B) | C) | D) |
|---|---|---|---|
| 1with open('abc.txt','r') f:<br>data=f.read() | with open('abc.txt') as f:<br>data=f.read() | with open('abc.txt','w') as f:<br>data=f.read() | with open('abc.txt') as f:<br>data=f.readline() |

Answer: B

## 159.

Assume that we are writing python code for some voting application.
You need to open the file voters_list.txt and add new voters info and print total data to the console?

```
1) with open('voters_list.txt','a+') as f:
2)    f.write('New voters info')
3)    #Line-1
4)    data=f.read()
5)    print(data)
```

Which Line should be inserted at Line-1 ?
A) f.seek(0)
B) f.flush()
C) f.begin()
D) f.close()

Answer: A

## 3.2 Construct and analyze code segments that perform console input and output operations

• Read input from console, print formatted text (string.format() method, f-String method), use command-line arguments

### 160.

You write the following code:

```
import datetime
d = datetime.datetime(2017, 4, 7)
print('{:%B-%d-%y}'.format(d))

num=1234567.890
print('{:, .4f}'.format(num))
```

You run the program.
What is the output?

A. 2017--April--07
   1,234,567.890
   Press any key to continue...
B. Apr--07--2017
   1,234,567,8900
   Press any key to continue...
C. April--07--17
   1,234,567.8900
   Press any key to continue...
D. April--07--17
   1234567.89
   Press any key to continue...

### 161.

Northwind Traders has hired you as an intern on the coding team that creates e-commerce applications.
You must write a script that asks the user for a value. The value must be used as a whole number in a calculation, even if the user enters a decimal value. You need to write the code to meet the requirements.
Which code segment should you use?

A. totalItems = int(input("How many items would you like?"))
B. totalItems = str(input("How many items would you like?"))
C. totalItems = float(input("How many items would you like?"))
D. totalItems = input("How many items would you like?")

### 162.

You develop a Python application for your company.
You need to accept input from the user and print that information to the user screen.
You have started with the following code. Line numbers are included for reference only.

```
01 print("What is your name?")
02
03 print(name)
```

Which code should you write at line 02?

A. name = input
B. input(name)
**C. name = input()**
D. input("name")

## 163.

If the user enters 12345 as input,Which of the following code will print 12346 to the console?

| | |
|---|---|
| A)<br>1) count=input('Enter count value:')<br>2) print(count+1) | C)<br>1) count=eval(input('Enter count value:'))<br>2) print(count+1) |
| B)<br>1) count=input('Enter count value:')<br>2) print(int(count)+1) | D)<br>1) count=int(input('Enter count value:'))<br>2) print(count+1) |

Answer: B,C,D

## 164.

You are creating a program that shows a congratulation message to employees on their service anniversary.
You need to calculate the number of years of service and print a congratulatory message.
You have written the following code. Line numbers are included for reference only.

```
01 start = input("How old were you on your start date?")
02 end = input("How old are you today?")
03
```

You need to complete the program.
Which code should you use at line 03?

A. print(“congratulations on “ + int(end - start) + “ years of service!”)
B. print(“congratulations on “ + (int(end) - int(start)) + “ years of service!”)
C. print(“congratulations on “ + str(end - start) + “ years of service!”)
D. print(”congratulations on “ + str(int(end) - int(start)) + “ years of service!”)

## 165.

You are an intern for Northwind Electric Cars. You must create a function that calculates the average velocity of their vehicles on a 1320 foot (1/4 mile) track. The output must be as precise as possible.
How should you complete the code? To answer, select the appropriate code segments in the answer area

```
#Speed calculator
distance = [1]  (input(”Enter the distance traveled in feet”))
distance_miles = distance/5280     #convert to miles
time = [2]  (input(”Enter the time elapsed in seconds”))
time_hours = time/3600             #convert to hours
velocity = distance_miles/time_hours
print("The average velocity is : ", velocity, " miles/hour")
```

[1]: int, float, str

[2]: int, float, str

## 166.

You are intern for XYZ Cars Company.You have to create a function that calculates the average velocity of vehicle on a 2640 foot(1/2 mile)track. Consider the python code:

```
1) distance=xxx(input('Enter the distance travelled in feet:')) #Line-1
2) distance_miles=distance/5280
3) time=yyy(input('Enter the time elapsed in seconds:')) #Line-2
4) time_hours=time/3600
5) velocity=distance_miles/time_hours
6) print('The average Velocity:',velocity,'miles/hour')
```

To generate most precise output, which modifications should be done at Line-1 and atLine-2.
A) xxx should be replaced with float and yyy should be replaced with float
B) xxx should be replaced with float and yyy should be replaced with int
C) xxx should be replaced with int and yyy should be replaced with float
D) xxx should be replaced with int and yyy should be replaced with int

Answer: A

## 167.

You are creating an ecommerce script that accepts input from the user and outputs the data in a comma delimited format. You write the following lines of code to accept input:

```
item = input(”Enter the item name: ”)
sales = input(“Enter the quantity: ”)
```

The output must meet the following requirements:

- Strings must be enclosed inside of double-quotes
- Numbers must not be enclosed in quotes or other characters
- Each item must be separated with a comma

You need to complete the code to meet the requirements. Which three code segments should you use? Each correct answer presents a complete solution. Choose three.

A. print(’”{0}”,{1}’.format(item, sales))
B. print(item + ‘,’ + sales)
C. print(’”’ + item + ‘”,‘ + sales)
D. print(”{0},{1}”.format(item, sales))
E. print(’”%s”, %s’ % (item, sales))

## 168.

We are developing an app in which students will provide college name and city as input. If the student provides college name as DURGASOFT and city as Hyderabad, then our application has to provide the following greeting message. Welcome to DURGASOFT in Hyderabad Which of the following code can be used for this requirement?

| | |
|---|---|
| A)<br>1) college_name=input('Enter Your College Name:')<br>2) city=input('Enter Your City:')<br>3) print('Welcome to {} in {}'.format(college_name,city))<br><br>B)<br>1) college_name=read('Enter Your College Name:')<br>2) city=read('Enter Your City:')<br>3) print('Welcome to {} in {}'.format(college_name,city)) | C)<br>1) college_name=eval('Enter Your College Name:')<br>2) city=eval('Enter Your City:')<br>3) print('Welcome to {} in {}'.format(college_name,city))<br><br>D)<br>1) college_name=str('Enter Your College Name:')<br>2) city=str('Enter Your City:')<br>3) print('Welcome to {} in {}'.format(college_name,city)) |

Answer: A

## 169.

You are writing a function to read a data file and print the results as a formatted table. The data file contains information about fruit. Each record contains the name of the fruit the weight and the price.
You need to print the data so that it looks like the following sample:

Oranges 5.6 1.33
Apples 2.0 0.54
Grapes 10.2 10.96

Specifically, the print out must meet the following requirements:

- The fruit name must print left-aligned in a column 10 spaces wide.
- The weight must print right-aligned in a column 5 spaces wide with up to one digit after the decimal point.
- The price must print right-aligned in a column 7 spaces wide with up to two digits after the decimal point.

You write the following code. Line numbers included are for reference only.

```
1. def print_table(file):
2.   data = open(file,'r')
3.   for record in data:
4.      fields = record.split(",")
5.
```

How should you complete line 05? To answer, drag the appropriate code segments to the correct location. Each code segment may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
Code Segments

| | | | |
|---|---|---|---|
| print(" | print(" | print(" | print(" |
| {10:0} | {10:0} | {10:0} | {10:0} |
| {5:1f} | {5:1f} | {5:1f} | {5:1f} |
| {7:2f} | {7:2f} | {7:2f} | {7:2f} |
| {2:7.2f} | {2:7.2f} | {2:7.2f} | {2:7.2f} |
| {1:5.1f} | {1:5.1f} | {1:5.1f} | {1:5.1f} |
| {0:10} | {0:10} | {0:10} | {0:10} |

".format(fields[0], eval(fields[1]), eval(fields[2])))

Answer Area

print("{:<10} {:>5.1f} {:>7.2f}".format(fields[0], eval(fields[1]), eval(fields[2]))

## 170.

You are writing a Python program to validate employee numbers.
The employee number must have the format ddd-dd-dddd and consist only of numbers and dashes. The program must print True if the format is correct and print False if the format is incorrect.
How should you complete the code? To answer, select the appropriate code segmentsin the answer area.

```
[1]
      parts = ""
[2]
       [3]
   employee_number = input('Enter employee number (ddd-dd-dddd): ")
   parts = employee_number. split(' -')
   if len(parts) == 3:
      if len(parts[0]) == 2 and len(parts[1]) == 3 and len(parts[2]) == 4:
         if parts[0].isdigit() and parts[1].isdigit() and parts[2].isdigit():
                        [4]
   print(valid)
```

[1]:
- Employee_number = ""
- Employee_number = "sentinel"

[2]:
- while employee_number != "":
- while employee_number != "sentinel":

[3]:
- valid = False
- valid = True

[4]:
- valid = False
- valid = True

## 171

You are writing a Python program to validate employee numbers. The employee number must have the format dd-ddd-dddd and consists of only numbers and dashes.The program must print True if the format is correct,otherwise print False.

```
1) employee_number=input('Enter Your Employee Number(dd-ddd-dddd):')
2) parts=employee_number.split('-')
3) valid=False
4) if len(parts) == 3:
5) if len(parts[0])==2 and len(parts[1])==3 and len(parts[2])==4:
6) if parts[0].isdigit() and parts[1].isdigit() and parts[2].isdigit():
7) valid=True
8) print(valid)
```

Which of the following is True about this code?

A) It will throw error because misuse of split() method
B) It will throw error because misuse of isDigit() method
C) There is no error but it won't fulfll our requirement.
D) No changes are required for this code and it can fulfill requirement.

Answer: D

## 172

What will be the output of the following program?

```
01 x="ITVEDANT"
02 print("%20s",x)
```

A) 20 blank spaces after “ITVEDANT”
B) 20 blank spaces before “ITVEDANT”
C) None of these.
D) –

## 173

What will be the output of the following python program?

```
01 d='{a}{b}{a}'.format(a='hello',b='world')
02 print(d)
```

A) hello world
B) hello world hello
C) helloworldhello
D) hello hello world

## 174

Consider the following statements:

```
1. print('V:{:.2f}'.format(123.45678)) will print to the console V:123.46
2. print('V:{:.2f}'.format(123.4)) will print to the console V:123.40
3. print('V:{:8.2f}'.format(1.45678)) will print to the console V:1.46
4. print('V:{:08.2f}'.format(1.45678)) will print to the console V:00001.46
```

Which if the above statements are True?

A) only 1 and 2
B) only 1 and 3
C) only 2 and 4
D) 1, 2, 3 and 4

## 4. Code Documentation and Structure

### 4.1 Document code segments

• Use indentation, white space, comments, and docstrings; generate documentation by using pydoc

### 175.

You develop a application for your company. You want to add notes to your code so other team members will understand it. What should you do?

A. Place the notes after the last line of code separated by a blank line.
B. Place the notes inside of parentheses on any line.
C. Place the notes after the # sign on any line.
D. Place the notes before the first line of code separated by a blank line.

Answer: C

### 176.

You are writing an application that uses the sqrt function. The program must reference the function using the name squareRoot. You need to import the function. Which code segment should you use?

A. from math import sqrt as squareRoot
B. from math.sqrt as squareRoot
C. import math.sqrt as squareRoot
D. import sqrt from math as squareRoot

Answer: A

### 177.

Woodgrove Bank is migrating their legacy bank transaction code to Python. You have been hired to document the migrated code. Which documentation syntax is correct?

| | |
|---|---|
| A.<br>```Returns the current balance of the bank account<br>  def get_balance():<br>  return balance``` | B.<br>```def get_balance():<br>  /*Returns the current balance of the bank account*/<br>  return balance``` |
| C.<br>```//Returns the current balance of the bank account<br>  def get_balance():<br>  return balance``` | D.<br>```def get_balance():<br>  #Returns the current balance of the bank account<br>  return balance``` |

### 178.

You create a function to calculate the power of a number. You need to ensure that the function ¡s documented with comments. You create the following code. Line numbers are included for reference only.

```
01 # The calc_power function calculates exponents
02 # x is the base
03 # y is the exponent
04 # The value of x raised to the y power is returned
05 def calc_power(x, y):
06   comment = "#Return the value"
07   return x**y 4 raise x to the y power
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

| | Yes | No |
|---|---|---|
| Lines 01 through 04 will be ignored for syntax checking. | ○ | ○ |
| The pound sign (#) is optional for lines 02 and 03. | ○ | ○ |
| The string in line 06 will be interpreted as a comment. | ○ | ○ |
| Line 07 contains an inline comment. | ○ | ○ |

## 4.2 Construct and analyze code segments that include function definitions

• Call signatures, default values, return, def, pass

### 179.

Adventure Works Cycles is creating a program that allows customers to log the number of miles biked. The program will send messages based on how many miles the customer logs. You create the following code. Line numbers are included for reference only.

```
01
02   name = input("What is your name? ")
03   return name
04
05   calories = miles * calories_per_mile
06   return calories
07 distance = int(input("How many miles did you bike this week? "))
08 burn_rate = 50
09 biker = get_name()
10 calories_burned = calc_calories(distance, burn_rate)
11 print(biker, ", you burned about", calories_burned, "calories.")
```

You need to define the two required functions. Which code segments should you use for line 01 and line 04? Each correct answer presents part of the solution. Choose two.

A. 01 def get_name():
B. 01 def get_name(biker):
C. 01 def get_name(name):
D. 04 def calc_calories():
E. 04 def calc_calories(miles, burn_rate):
F. 04 def calc_calories(miles, calories_per_mile)

### 180.

You are writing a function that increments the player score in a game.
The function has the following requirements:

- If no value is specified for points, then points start at one
- If bonus is True, then points must be doubled

You write the following code. Line numbers are included for reference only.

```
01 def increment_score(score, bonus, points):
02   if bonus == True:
03      points = points * 2
04   score = score + points
05   return score
06 points = 5
07 score = 10
08 new_score = increment_score(score, True, points)
```

For each of the following statements, select Yes if the statement is true.
Otherwise, select No.

| | Yes | No |
|---|---|---|
| To meet the requirements, line 01 must be changed to the following: def increnient_score(score, bonus, points = 1): | ○ | ○ |
| Once any parameter is defined with a default value, any parameters to the right must also be defined with default values. | ○ | ○ |
| If the function is called with only two parameters, the value of the third parameter will be None. | ○ | ○ |
| Line 03 will also modify the value of the variable points declared at line 06. | ○ | ○ |

## 181.

You are writing a function in Python that must meet the following requirements:

- The function accepts a list and a string as parameters.
- The function must search for the string in the list.
- If the string is found in the list, the function must print a message indicating that the string was found and then stop iterating through the list.
- If the string is not found, the function must print a message indicating that the string was not found in the list.

In which order should you arrange the code segments to develop the solution? To answer, move all code segments from the list of code segments to the answer area and arrange them in the correct order.

```
A -    for i in range(len(items)):
```

```
B -    if items[i]==term:
           print("{0} was found in the list.".format(term))
```

```
C - break
```

```
D - def search(items,term):
```

```
E -
    else:
        print("{0} was not found in the list." .format(term))
```

A) D,A,B,C,E
B) D,B,A,C,E
C) B,D,A,C,E
D) D,A,B,E,C

## 182.

You are developing a Python application for an online game. You need to create a function that meets the following criteria:

- The function is named update_score
- The function receives the current score and a value
- The function adds the value to the current score
- The function returns the new score

How should you complete the code? To answer, select the appropriate code segments in the answer area:

```
[1] [ ] [2] [ ]
    current += value
    [3] [ ]
```

[1]:
- update_score
- def update_score
- return update_score

[2]:
- (current, value):
- (current, value)
- ():
- ()

[3]:
- pass current
- return current
- return
- pass

## 183.

You are developing a Python application for online game. You need to create a function that meets the following criteria:

- The function is named update_score
- The function receives the current score and a value.
- The function adds the value to the current score.
- The function returns the new score.

Which of the following is valid function to fulfill this requirement?

| A | B | C | D |
|---|---|---|---|
| update_score(score,value):<br>new_score=score+value<br>return new_score | def update_score(score,value):<br>new_score=score+value<br>return new_score | def update_score(score,value):<br>new_score=score+value<br>pass new_score | def update_score():<br>new_score=score+value<br>return new_score |

Answer: B

## 184.

A coworker wrote a program that inputs names into a database. Unfortunately, the program reversed the letters in each name. You need to write a Python function that outputs the characters in a name in the correct order
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
#Function reverses characters in a string.
#returns new string in reversed order.
def reverse_name(backwards_name):
   forward_name = ''
      for index in [1]
         forward_name += [2]
      return forward_name
print(reverse_name("leinad"))   #test_ case
```

[1]:
- backwards_name:
- len(backwards_name:)
- range(0, len(backwards_name),-1):
- range(len(backwards_name)-1,-1,-1):

[2]:
- backwards_name[index-1]
- backwards_name[len(forward_name)-1]
- backwards_name[len(backward_name)- len(forwardname)
- backwards_name[index]

## 185.

You are writing a function to perform safe division. You need to ensure that a denominator and numerator are passed to the function and that the denominator is not zero.
How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
def safe_divide(numerator, denominator):
   [1]
      print("A required value is missing.")
   [2]
      print("The denominator is zero.")
   else:
      return numerator I denominator
```

[1]:
- if numerator is None or denominator is None:
- if numerator is None and denominator is None:
- if numerator = None or denominator = None:
- if numerator = None and denominator = None:

[2]:
- elif denominator == 0:
- elif denominator = 0:
- elif denominator != 0:
- elif denominator in 0:

## 186.

Wingtip Toys is creating an interactive Times Table Helper program intended for elementary school children. You need to complete a function that computes and displays all multiplication table combinations from 2 to 12. How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
#Displays times tables 2 - 12
def times_tables():
   [1]
      [2]
         print( row*col, end=" ")
      print()
#main
times_tables()
```

[1]:
- for col in range(13):
- for col in range(2, 12, 1):
- for col in range(2, 13):
- for col in range(12):

[2]:
- for row in range(13):
- for row in range(2, 12, 1):
- for row in range(2, 13):
- for row in range(12):

## 187.

Consider the following code:

```
def get_score(total=0,valid=0):
   result=int(valid)/int(total)
   return result
```

For which of the function calls we will get Error?
A) score = get_score('40', '4')
B) score = get_score(0, 10)
C) score = get_score(40, 4)
D) score = get_score(40)

Answer: B

## 188.

Consider the code

```
1) def get_names():
2) names=['Sunny','Bunny','Chinny','Vinny','Pinny']
3) return names[2:]
4)
5) def update_names(elements):
6) new_names=[]
7) for name in elements:
8) new_names.append(name[:3].upper())
9) return new_names
10)
11) print(update_names(get_names()))
```

What is the result?
A) ['CHI', 'VIN', 'PIN']
B) ['VIN', 'PIN']
C) ['CH', 'VI', 'PI']
D) ['SU', 'BU']
Answer: A

## 189.

Consider the following code

```
1) def my_list(x):
2) lst.append(a)
3) return lst
4)
5) my_list('chicken')
6) my_list('mutton')
7) print(my_list('fish'))
```

to print the following to the console

```
['chicken','mutton','fish']
```

x should be replaced with
A) a,lst=[]
B) a,lst=()
C) a,lst={}
D) a,lst=None

Answer: A

## 190.

Consider the following code:

```
1) def f1(x=0,y=0):
2) return x+y
```

Which of the following method calls are valid?
A) f1()
B) f1('10','20')
C) f1(10)
D) f1('10')

Answer: A,B,C

## 191.

Consider the following code:

```
1) def f1(x=0,y=0):
2) return x*y
```

Which of the following method calls are valid?
A) f1()
B) f1('10','20')
C) f1(10)
D) f1('10')

Answer: A,C,D

## 192.

Consider the following code:

```
1) numbers=[100,20,10,70,50,60,40,30,90,80]
2) #Insert Code Here
3) print('The highest Number:{} and Least Number:{}'.format(high,low))
```

Which of the following code should be inserted to print Highest Number as 100 and Least Number as 10

| | |
|---|---|
| A)<br>1) def find_numbers():<br>2) numbers.sort()<br>3) return numbers[0],numbers[-1]<br>4) low,high=find_numbers() | C)<br>1) def find_numbers():<br>2) numbers.sort()<br>3) return numbers[0],numbers[-1]<br>4) low=find_numbers()<br>5) high=find_numbers() |
| B)<br>1) def find_numbers():<br>2) numbers.sort()<br>3) return numbers[0],numbers[len(numbers)]<br>4) low,high=find_numbers() | D)<br>1) def find_numbers():<br>2) numbers.sort()<br>3) return numbers[2],numbers[0]<br>4) low,high=find_numbers() |

Answer: A

## 193.

Consider the code:

```
1) numbers=[100,20,10,70,50,60,40,30,90,80]
2)
3) def find_numbers():
4)   numbers.sort()
5)   return numbers[0],numbers[-1]
6)
7) low=find_numbers()
8) high=find_numbers()
9) #Line-1
```

To print 10 100 to the console which of the following code we have to take at Line-1

A) print(low,high)
B) print(low[0],high[-1])
C) print(low[-1],high[0])
D) print(low[2],high[0])

Answer: B

## 194.

Consider the code:

```
1) def calculate(amount=6,factor=3):
2)   if amount>6:
3)     return amount*factor
4)   else:
5)     return amount*factor*2
```

Which of the following function calls returns 30

A) calculate()
B) calculate(10)
C) calculate(5,2)
D) calculate(1)

Answer: B

## 195.

Consider the following code

```
1) def fib_seq(n):
2)   if n==0:
3)     return 0
4)   elif n==1:
5)     return 1
6)   else:
7)     return fib_seq(n-1)+fib_seq(n-2)
8) for i in range(7):
9)   print(fib_seq(i),end=',')
```

What is the result?
A) 0,1,1,2,3,5,8,
B) 0,1,2,4,8,16,32,
C) 0,1,0,2,0,4,0,
D) None of these

Answer: A

## 5. Troubleshooting and Error Handling

### 5.1 Analyze, detect, and fix code segments that have errors

• Syntax errors, logic errors, runtime errors

### 196.

Tailspin Toys uses Python to control its new toy Happy Clown. The program has errors that cause the clown to run around in an infinite circle. You have been hired to help debug the following Happy Clown code. Line numbers are included for reference only.

```
01 import math
02 #default motion for happy clown
03 power = True
04 move = 0
05 while(power):
06   if move == 0:
07     turnValue = math.pi/move
08     move+=5
09   else:
10     turnValue = 0
11     move = 0
```

A. Line 05 has a syntax error because it should read (power == True).
B. Line 08 has a syntax error because + = is an invalid statement.
C. Line 07 causes a runtime error due to division by zero.
D. Line 05 causes a runtime error because the expression is incomplete.

### 197.

| Consider the code | What is the result? |
|---|---|
| `def area(b,w):`<br>`  return B*w`<br>`print(area(10,20))` | A) AttributeError will be raised at runtime<br>B) IndentationError will be raised at runtime<br>C) NameError will be raised at runtime<br>D) 200<br><br>Answer: C |

### 198.

Consider the code:

```
1) data=[]
2) def get_data():
3)   for i in range(1,5):
4)     marks=input('Enter Marks:')
5)     data.append(marks)
6)
7) def get_avg():
8)   sum=0
9)   for mark in data:
10)    sum += mark
11)  return sum/len(data)
12) get_data()
13) print(get_avg())
```

For the input: 10,20,30,40 what is the result?
A) 25
B) 25.0
C) NameError is thrown at runtime
D) TypeError is thrown at runtime

Answer: D

## 199.

Consider the code:

```
1) a=10
2) b=0
3) try:
4)   print(a/b)
```

Which of the following except block print the name of exception which is raised,i.e exception class name?

| | |
|---|---|
| A)<br>1) except ZeroDivisionError as e:<br>2) print('Exception Type:',e.__class__.__name__) | C)<br>1) except ZeroDivisionError as e:<br>2) print('Exception Type:',e) |
| B)<br>1) except ZeroDivisionError as e:<br>2) print('Exception Type:',type(e).__name__) | D) All of these |

Answers : A,B

## 200.

Which of the following is valid way of creating our own custom exception?

| A) | B) | C) | D) |
|---|---|---|---|
| class MyException:<br>pass | class MyException():<br>pass | class MyException(Exception):<br>pass | It is not possible to define custom exceptions in python |

Answer: C

## 201.

Consider the code

1) x=int(input('Enter First Number:'))
2) y=int(input('Enter Second Number:'))
3) try:
4) print(x/y)

Which of the following is valid except block that handles both ZeroDivisionError and ValueError

| | |
|---|---|
| A)<br>except(ZeroDivisionError,ValueError) from e:<br>print(e) | C)<br>except(ZeroDivisionError \| ValueError) as e:<br>print(e) |
| B)<br>except(ZeroDivisionError,ValueError) as e:<br>print(e) | D)<br>except(ZeroDivisionError, ValueError as e):<br>print(e) |

Answer: B

## 5.2 Analyze and construct code segments that handle exceptions

• try, except, else, finally, raise

### 202.

You write the following code:

```
import sys
try:
    file_in = open("in.txt", 'r')
    file_out = open("out.txt", 'w+')
except IOError:
    print('cannot open', file_name)
else:
    i = 1
    for line in file_in:
        print(line. rstrip())
        file_out.write("line " + str(i) + ": " + line)
        i = i + 1
    file_in.close()
    file_out.close ()
```

The out.txt file does not exist. You run the code. The code will execute without error. Review the underlined text. If it makes the statement correct, select "No change is needed." If the statement is incorrect, select the answer choice that makes the statement correct.

A. No change is needed.
B. The code runs but generates a logic error.
C. The code will generate a runtime error.
D. The code will generate a syntax error.

### 203.

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

| | Yes | No |
|---|---|---|
| A try statement can have one or more except clauses. | ○ | ○ |
| A try statement can have a finally clause without an except clause. | ○ | ○ |
| A try statement can have a finally clause and an except clause. | ○ | ○ |
| A try statement can have one or more finally clauses. | ○ | ○ |

### 204.

You are creating a program that accepts user input. The program must cast the input into an integer. You must properly handle the error if the code cannot cast the input to an integer.

How should you complete the code? To answer, select the appropriate code segments in the answer area.

```
while True:
    [1]
        x = int(input("Please enter a number: "))
        break
    [2]  ValueError:
        print("Not a valid number. Try again...")
```

[1]: [2]:

try:
else:
except:
raise:
finally:

## 205.
When will the else part of try-except-else be executed?

A) always
B) when no exception occurs
C) when error exception
D) when an exception occurs in to except block

## 206.
Is the following Python code valid?

```
try :
   Try block
except:
   Except block
finally :
   Finally Block
```

A) no, there is no such thing as finally
B) no, finally cannot be used with except
C) no, finally must come before except
D) yes

## 207.
What will be the output of the following python code?

```
def foo():
  try:
    return 1
  finally:
    return 2
k = foo()
print(k)
```

A) 1
B) 2
C) 3
D) error, there is more than one return statement in a single try-finally block

## 208.
Consider the code

```
def f1():
    try:
        return 1
    finally:
        return 2

x = f1()
print(x)
```

What is the result?
A) 1
B) 2
C) print both 1 and 2
D) Error, because more than one return statement is not allowed

## 209.

Consider the code:

```
f = open('abc.txt')
print(f.read())
f.close()
```

We required to add exception handling code to handle FileNotFoundError.
We of the following is appropriate code for this requirement?

| Code A: | Code B: | Code C: |
|---|---|---|
| f = None<br>try:<br>  f = open('abc.txt')<br>except FileNotFoundError:<br>  print("File does not exist")<br>else:<br>  print(f.read())<br>finally:<br>  if f is not None:<br>  f.close() | f = None<br>try:<br>  f = open('abc.txt')<br>except FileNotFoundException:<br>  print("File does not exist")<br>else:<br>  print(f.read())<br>finally:<br>  if f is not None:<br>  f.close() | f = None<br>try:<br>  f = open('abc.txt')<br>else:<br>  print(f.read())<br>except FileNotFoundError:<br>  print("File does not exist")<br>finally:<br>  if f is not None:<br>  f.close() |

A) Code A
B) Code B
C) Code C
D) None of these

Answer: A

## 210.

Consider the code:

```
import sys

try:
    file_in = open('in.txt', 'r')
    file_out = open('out.txt', 'w+')
except IOError:
    print('Cannot open file:', 'in.txt')
else:
    i = 1
    for line in file_in:
        print(line.rstrip())
        file_out.write(str(i) + ": " + line)
        i += 1
    file_in.close()
    file_out.close()
```

Assume that in.txt file is available but out.txt file does not exist.
Which of the following is true about this code?

A) The code runs, but generates logical error
B) This program will copy data from in.txt to out.txt
C) The code will generates a runtime error
D) The code will generate a syntax error

Answer: B

## 211.

Consider the file abc.txt has the following content:

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

We have to write python code to read total data and print to the console.

```
try:
  f=open('abc.txt','r')
  //Line-1
except:
  print('Unable to open the file')
  print(data)
```

Which code should be inserted at Line-1 to meet the given requirement?

A) data = f.readlines()
B) data = f.readline()
C) data = f.read()
D) data = f.load()

Answer: C

## 212.

The base class for all exceptions in python is:

A) Exception
B) ExceptioNBase
C) BaseException
D) ArithmeticError

## 213.

Consider the code:

```
a=10
b=20
c='30'
result=a+b+c
```

What is the result?

A) 102030
B) 3030
C) TypeError
D) ArithmeticError

Answer: C

## 214.

Consider the code:

```
1) try:
2)   print('try')
3) except:
4)   print('except')
5) else:
6)   print('else')
7) finally:
8)   print('finally')
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| try<br>except<br>else<br>finally | try<br>else<br>finally | try<br>except<br>finally | try<br>finally |

Answer:B

## 215.

Consider the code

```
1) try:
2)   print('try')
3)   print(10/0)
4) except:
5)   print('except')
6) else:
7)   print('else')
8) finally:
9)   print('finally')
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| try<br>except<br>else<br>finally | try<br>else<br>finally | try<br>except<br>finally | try<br>finally |

Answer:C

## 216.

Consider the code

```
1) try:
2)   print('try')
3)   print(10/0)
4) else:
5)   print('else')
6) except:
7)   print('except')
8) finally:
9)   print('finally')
```

What is the result?

| A) | B) | C) | D) |
|---|---|---|---|
| try<br>except<br>else<br>finally | try<br>else<br>finally | try<br>except<br>finally | SyntaxError: invalid syntax |

Answer:D

## 217.

Consider the code:

```
1) prices=[30.5,'40.5',10.5]
2) total=0
3) for price in prices:
4)    total += price
5) print(total)
```

While executing this code we are getting the following error:

```
Traceback (most recent call last):
File "test.py", line 4, in <module>
total += price
TypeError: unsupported operand type(s) for +=: 'float' and 'str'
```

Which of the following code should be used to fix this error?
A) total += str(price)
B) total += int(price)
C) total += float(price)
D) total = total+price

Answer: C

## 218.

Consider the code:

```
1) prices=[10,'20',30,'40']
2) total=0
3) for price in prices:
4)   total +=price
5) print(total)
```

While executing this code we are getting the following error

```
Traceback (most recent call last):
File "test.py", line 4, in <module>
total +=price
TypeError: unsupported operand type(s) for +=: 'int' and 'str'
```

By using which of the following code segments we can fix this problem(Choose 2)?
A) total += str(price)
B) total += int(price)
C) total += float(price)
D) total = total+price

Answer: B and C

## 219.

Consider the code

```
1) courses={1:'Java',2:'Scala',3:'Python'}
2) for i in range(1,5):
3) print(courses[i])
```

While executing this code we are getting the following error:

```
Traceback (most recent call last):
File "test.py", line 3, in <module>
print(courses[i])
KeyError: 4
```

By using which of the following code segments we can fix this problem ?

| A | B | C |
|---|---|---|
| courses={1:'Java',2:'Scala',3:'Python'}<br>for i in range(1,5):<br>  if i in courses:<br>  print(courses[i]) | courses={1:'Java',2:'Scala',3:'Python'}<br>for i in courses:<br>  print(courses[i]) | courses={1:'Java',2:'Scala',3:'Python'}<br>for i in range(1,4):<br>  print(courses[i]) |
| D | | |
| All of these | | |

Answer: D

## 5.3 Perform unit testing

• Unittest, functions, methods, and assert methods (assertIsInstance, assertEqual, assertTrue, assertIs, assertIn)

### 220.

What will be the output of the following python code?

```
x =30
y = 10
assert x > y ,'x is smaller than y'
```

A) Assertion Error
B) 10 8
C) No output
D) 108

### 221.

Which type of exception will be raised if we are trying to call a method on the inappropriate object?

A) IndexError
B) TypeError
C) AttributeError
D) None of these

Answer: C

### 222.

Consider the code:

```
f=open('abc.txt')
f.readall()
```

Which exception will be raised?

A) AttributeError
B) EOFError
C) SystemError
D) SyntaxError

Answer: A

## 6. Operations using Modules and Tools

### 6.1 Perform basic file system and command-line operations by using built-in modules

• io, os, os.path, sys (importing modules, using modules to open, read, and check existence of files, command-line arguments)

### 223.

You write a function that reads a data file and prints each line of the file. You write the following code. Line numbers are included for reference only.

```
01 def read_file(file):
02     line = None
03     if os.path.isfile(file):
04         data = open(file, 'r')
05         for line in data:
06             print(line)
```

When you run the program, you receive an error on line 03.
What is causing the error?

A. The **path** method does not exist in the **os** object.
B. The **isfile** method does not exist in the **path** object.
C. You need to import the **os** library.
D. The **isfile** method does not accept one parameter.

### 224.

What is the type of sys.argv?
A) set
B) list
C) tuple
D) string

### 225.

Which of the following isn't true about main modules?
A) When a python file is directly executed, it is considered main module of a program
B) Main modules may import any number of modules
C) Special name given to main modules is: _main_
D) Other main modules can import main modules

### 226.

From sys module, by using which variable we can access command line arguments?
A) argv
B) argsv
C) args
D) arguments

### 227.

Given the command invocation:

```
python tests.py Itvedant
```

Which of the following code prints 'Itvedant' to the console?
A) from sys import argv; print(argv[1])
B) from sys import argv; print(argv[0])
C) from sys import args; print(args[0])
D) from sys import args; print(args[1])

Answer: A

## 228.
Consider the code:

```
1) from sys import argv
2) print(argv[0])
```

and given the command invocation:

```
py test.py DURGASOFT
```

What is the result?
A) DURGASOFT
B) test.py
C) IndexError will be thrown at runtime
D) ImportError will be thrown at runtime

Answer: B

## 229.
Consider the code:

```
1) from sys import argv
2) print(argv[1]+argv[2])
```

and given the command invocation:

```
py test.py 10 20
```

What is the result?
A) 30
B) 1020
C) IndexError will be thrown at runtime
D) ImportError will be thrown at runtime
Answer: B

## 230.
Consider the code :

```
1) from sys import argv
2) sum=0
3) for i in range(2,len(argv)):
4) sum += float(argv[i])
5) print("The Average for {0} is {1:.2f}".format(argv[1],sum/(len(argv)-2)))
```

Which of the following command invocations will generate the output:

```
The Average for Durga is 20.00
```

A) py test.py Durga 10 20 30
B) py test.py Durga 10 20
C) py test.py Durga 10
D) py test.py 20

Answer: A

## 6.2 Solve complex computing problems by using built-in modules

• Math (fabs, ceil, floor, trunc, fmod, frexp, nan, isnan, sqrt, isqrt, pow, pi) datetime (now, strftime, weekday), random (randrange, randint, random, shuffle, choice, sample)

### 231.

You need to write code that generates a random float with a minimum value of 0.0 and a maximum value of 1.0. Which statement should you use?

A. random.randrange()
B. random.randrange(0.0, 1.0)
C. random.random()
D. random.randint(0, 1)

### 232.

You are writing code that generates a random integer with a minimum value of 5 and a maximum value of 11. Which two functions should you use? Each correct answer presents a complete solution. Choose two.

A. random.randint(5, 11)
B. random.randrange(5, 12, 1)
C. random.randint(5, 12)
D. random.randrange(5, 11, 1)

### 233.

You work on a team that is developing a game for AdventureWorks. You need to write code that generates a random number that meets the following requirements:

- The number is a multiple of 5.
- The lowest number is 5.
- The highest number is 100.

Which two code segments will meet the requirements? Each correct answer presents a complete solution. Choose two.

A. from random import randrange
print(randrange(5, 100, 5))
B. from random import randint
print(randint(1, 20) * 5)
C. from random import randint
print(randint(0, 20) * 5)
D. from random import randrange
print(randrange(0, 100, 5))

### 234.

You are creating a function that manipulates a number. The function has the following requirements:

- A float is passed into the function
- The function must take the absolute value of the float
- Any decimal points after the integer must be removed

Which two math functions should you use? Each correct answer is part of the solution. Choose two.

A. math.ceil(x)
B. math.fmod(x)
C. math.floor(x)
D. math.frexp(x)
E. math.fabs(x)

Answer: E and C

## 235.
What will be the output of the following Python code?

```
from math import factorial
print(math.factorial(5))
```

A) 120
B) Nothing is printed
C) Error, method factorial doesn't exist in math module
D) Error, the statement should be: print(factorial(5))

## 236.
You are writing an application that uses the pow() function. The program must reference the function using the name power. You need to import the function.
Which code segment should you use?

A) import math.pow as power
B) import pow from math as power
C) from math import pow as power
D) from math.pow as power.

## 237.
What is returned by the math.ceil(10.4)?
A) 11
B) 10
C) 11.0
D) 10.0

## 238.
What will be the output of the following Python code if the system date is 23rd September, 2020?

```
t=datetime.date.today()
print(t.month())
```

A) September
B) Sept
C) 09
D) 9

## 239.
Consider the code:

```
1) import math
2) l =[str(round(math.pi)) for i in range (1, 6)]
3) print(l)
```

What is the result?

A) ['3', '3', '3', '3', '3']
B) ['3', '3', '3', '3', '3','3']
C) ['1', '2', '3', '4', '5']
D) ['1', '2', '3', '4', '5','6']
Answer: A

## 240.

Consider the code:

```
1) import random
2) fruits=['Apple','Mango','Orange','Lemon']
```

Which of the following will print some random value from the list?

A) print(random.sample(fruits))
B) print(random.sample(fruits,3)[0])
C) print(random.choice(fruits))
D) print(random.choice(fruits)[0])

Answer: B and C

## 241.

We are developing an application for the client requirement.As the part of that we have to create a list of 7 random integers between 1 and 7 inclusive. Which of the following code should be used?

| | |
|---|---|
| A)<br>1) import random<br>2) randints=[random.randint(1,7) for i in range(1,8)] | C)<br>1) import random<br>2) randints=random.randrange(1,7) |
| B)<br>1) import random<br>2) randints=[random.randint(1,8) for i in range(1,8)] | D)<br>1) import random<br>2) randints=random.randint(1,7) |

Answer: A

## 242.

Consider the code:

```
1) import random
2) fruits=['Apple','Mango','Orange','Lemon']
3) random_list=[random.choice(fruits)[:2] for i in range(3)]
4) print(''.join(random_list))
```

Which of the following are possible outputs?

A) ApApAp
B) ApMaOr
C) LeMaOr
D) OrOraM

Answer: A,B,C

## 243.

We are developing a mathematical function to find area for the given circle. if r is the radius then area is : pi*r**2
Which of the following is valid function for this requirement?

| | |
|---|---|
| A)<br>1) import math<br>2) def find_area(r):<br>3) return math.pi*math.fmod(r,2) | B)<br>1) import math<br>2) def find_area(r):<br>3) return math.pi*math.fabs(r) |
| C)<br>1) import math<br>2) def find_area(r):<br>3) return math.pi*math.pow(r,2) | D) None of these |

Answer: C

## 244.
Consider the python code:

```
1) import random
2) print(int(random.random()*5))
```

Which of the following is true?

A) It will print a random int value from 0 to 5
B) It will print a random int value from 1 to 5
C) It will print a random int value from 0 to 5
D) It will print a random int value from 0 to 4
E) It will print 5

Answer: D

## 245.
Consider the code

```
1) import random
2) print(random.sample(range(10), 7))
```

Which of the following is valid?

A) It will print list of 10 unique random numbers from 0 to 6
B) It will print list of 7 unique random numbers from 0 to 9
C) It will print list of 7 unique random numbers from 0 to 10
D) It will print list of 7 unique random numbers from 1 to 10

Answer: B